Considers the dynamics of the load.

AD-Planner: Adaptive In-Flight Delivery Using a Quadrotor With a Suspended Payload

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***Abstract*—Aerial delivery is emerging as an effective approach in complex environments, offering rapid response and access to hard-to-reach areas. A quadrotor with a suspended payload system is a versatile solution for aerial delivery, combining mechanical simplicity with the ability to cast payloads into confined or moving targets. Despite the potential, achieving fully autonomous delivery across dynamic targets remains challenging due to accuracy, safety, and efficiency constraints. In this article, we present a systematic solution for autonomous delivery using a quadrotor with a suspended payload. Our approach addresses the complete delivery process, encompassing picking, transportation, and casting stages, and is designed to handle dynamic targets. To achieve accurate picking and casting, we introduce an error-bounded planning method that accounts for uncertainty. Moreover, we propose a compact optimization formulation that simultaneously considers safety, feasibility, and efficiency, thereby enabling seamless and agile delivery operations. To further improve computational efficiency and solution quality, we develop a hierarchical initialization strategy. The simulation and real-world experiments demonstrate the effectiveness of our method in generating smooth, feasible, and efficient trajectories for delivery scenarios.**

***Index Terms*—Aerial delivery, aerial system, application, aerial system, perception and autonomy, motion and path planning.**

I. Introduction

DELIVERY in complex environments, such as mountainous regions, disaster-stricken areas, and offshore platforms, presents significant challenges. Aerial transportation using quadrotors has emerged as an effective solution in these extreme conditions, owing to its flexibility and efficiency [1]. This manner enables access to hard-to-reach locations without being constrained by the terrain and offers a faster alternative to traditional ground-based methods.

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Among various UAV configurations, a quadrotor with a suspended payload system (referred to as quadrotor-payload system hereafter) has attracted attention due to its mechanical simplicity, agility, and high load capacity [2], [3]. A key advantage of this system is its ability to dynamically cast payloads, which is valuable in several scenarios. First, it enables delivery into confined or narrow spaces where direct access is restricted, such as throwing payloads through a window during a high-rise building fire rescue. Second, the system is well-suited for dynamic delivery onto moving targets like vehicles or assembly lines, ensuring effective operation in fast-paced environments.

In this article, we aim to harness the potential of quadrotor-payload systems for a fully autonomous delivery process, which consists of three stages: picking, transportation, and casting, as illustrated in Fig. 1. To maximize time efficiency, the system is designed to complete these stages in flight seamlessly and continuously, without any pauses. During this process, the quadrotor should accurately pick up the payload from a platform, transport it to the target location through complex environments, and cast it into the designated area. We assume that both the picking and casting targets may be moving during operation. Therefore, higher computational efficiency is necessary for onboard replanning. To achieve this, an *adaptive* planning method that explicitly considers accuracy, flexibility, and computational efficiency is essential to address the practical challenges analyzed as follows.

The first challenge is achieving *accurate* payload picking and casting in-flight. In the quadrotor-payload system, the payload is suspended by a cable, which introduces passive dynamics that make it more difficult to control compared to a standalone quadrotor, resulting in significant control errors. As a result, it becomes challenging to ensure that the payload's position and velocity at the pickup or release moment align precisely with the desired values. Meanwhile, the payload's state becomes uncontrollable once released. Any deviation in the release velocity is gradually amplified as the payload descends, leading to significant landing errors. Therefore, precisely casting the payload into the desired location is particularly difficult. In addition, dynamic targets for picking and casting introduce further uncertainties, making the task even more complex. Previous work [4] has often overlooked these error amplification effects, resulting in unreliable casting accuracy. To meet the accuracy requirement, these uncertainties should be explicitly modeled and constrained within an acceptable margin by the planner.

The second challenge is to conduct the entire delivery process *flexibly* while adhering to various safety and feasibility

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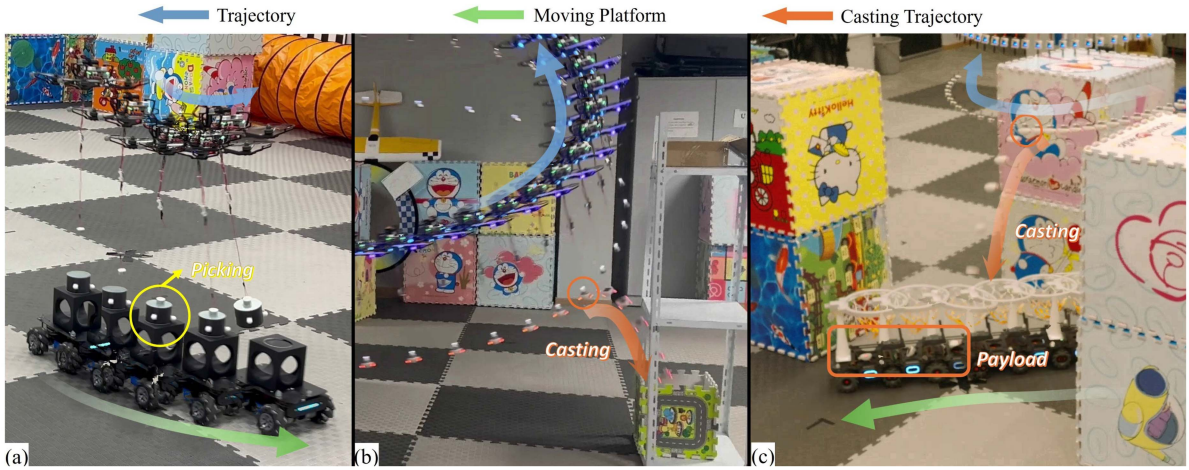


Fig. 1. Delivery using a quadrotor with a suspended payload in complex environments. The delivery involves three stages: picking, transportation, and casting. (a) Picking from a moving platform. (b) Casting to an occluded box. (c) Casting to a moving platform.

constraints. Previous studies [5] have relied on predefined waypoints to achieve picking and casting. However, using fixed picking and casting positions restricts the solution space for the trajectory, often leading to time-inefficient paths and potential failure in complex environments. Therefore, the planner should flexibly determine the suitable time for picking and casting, minimizing total flight duration and control efforts. In addition, ensuring safety throughout the operation is critical. In the casting stage, because the payload's state is uncontrollable after release, the casting trajectory should account for uncertainties in the payload's motion to avoid contact with obstacles in confined spaces. Moreover, the generated motions need to be dynamically feasible for the quadrotor-payload system. A flexible planner should consider all these factors and generate a trajectory that satisfies accuracy, safety, and feasibility while optimizing overall time efficiency.

Another challenge is *computational efficiency*. The optimization problem incorporating uncertainty and dynamic targets is inherently nonlinear and complex, making trajectory generation challenging with the limited computational resources onboard the quadrotor. The changing target states in dynamic environments necessitate the planner to rapidly adapt to new conditions. A high-quality initial guess is crucial to avoid local minima and accelerate convergence. However, a previous approach [6] initializes only the positions along the trajectory, which is sufficient for transportation tasks but inadequate for casting. For casting tasks, initialization of both position and velocity is critical. Different casting velocities mean different casting trajectories. Incorrect velocity initialization may trap the optimization in poor local minima and increase the computation time. Therefore, a comprehensive initialization strategy is necessary to enhance the solution quality and computational efficiency of the optimization problem.

To address these challenges, we propose a systematic solution for delivery using a quadrotor with a suspended payload. Our method is centered around error-bounded planning: we derive how bounded control errors propagate to pickup or casting error bounds and incorporate the resulting bounds into a compact trajectory optimization that spans the full delivery pipeline. Our approach integrates several key components to tackle the major challenges of *accuracy*, *flexibility*, and *computational efficiency*. To ensure accuracy, we employ robust optimization to certify worst-case performance during both casting and picking within prescribed bounds. A simple yet effective uncertainty model for the payload states is defined. This uncertainty model is then converted into deterministic constraints, allowing us to bound the casting and picking errors within a specified range. Moreover, our formulation accounts for the uncertainty associated with dynamic targets, facilitating interactions with moving platforms during both pickup and casting.

For flexibility, we present a compact optimization formulation that unifies picking, transportation, and casting in a single framework. Our formulation jointly optimizes all three stages, automatically determining the picking and casting timing based on environmental and task constraints. For enhanced safety, our method bounds the envelope of the payload's trajectory, guaranteeing obstacle avoidance even after release. This formulation allows the planner to generate a trajectory that meets accuracy, safety, and feasibility requirements while optimizing overall time efficiency.

Finally, we design a hierarchical initialization strategy to enhance solution quality and computational efficiency. This strategy features a warm-up optimization that accurately initializes both positions and velocities, which is critical for avoiding poor local minima in casting tasks and improving computational efficiency. We also introduce an iterative search strategy that effectively accommodates moving targets. This progressive strategy effectively navigates the complex solution space, yielding high-quality trajectories with high computational efficiency.

To evaluate the effectiveness of the proposed method, we conduct simulation and real-world experiments under selected static and dynamic-target conditions. The results show that the proposed method can deliver payloads to desired locations accurately and safely in these tested scenarios. To the best of authors' knowledge, this is the first work to address the complete

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delivery process seamlessly, including picking, transportation, and casting, using a quadrotor with a suspended payload. The contributions of our work can be summarized as follows.

1) An error-bounded planning approach that provides accuracy guarantees by explicitly modeling control errors and deriving picking and casting error bounds.
2) A compact optimization formulation that unifies picking, transportation, and casting into a single framework, which jointly optimizes the entire trajectory for accuracy, safety, feasibility, and time efficiency.
3) A hierarchical initialization strategy for optimization, which enhances the solution quality and computational efficiency for the nonlinear optimization problem.

II. Related Work

A. System Configuration for Aerial Transportation

Researchers have explored two major system configurations in aerial transportation using quadrotors: quadrotors equipped with suspended payloads [6], [7], [8], [9] and quadrotors integrated with robotic arms [10], [11], [12] or grippers [3], [13], [14], [15].

Quadrotors with robotic arms [10], [11], [12] offer higher degrees of freedom (DoFs), enabling more precise interactions with objects [16]. However, the introduced mechanical complexity and weight often degrade energy efficiency and agility. In response, Fishman et al. [13] developed a lightweight gripper, and Wu et al. [14] proposed a morphing design to reduce system weight. Despite these advancements, such systems rely on a rigid connection between the UAV and the payload. Consequently, the attitude dynamics of the quadrotor and the payload remain coupled, restricting the system's agility.

In contrast, quadrotors with suspended payloads feature a much simpler mechanical structure, avoiding the additional weight of robotic arms [6]. The suspended connection decouples the attitude dynamics between the quadrotor and the payload, enhancing the system's agility [7]. This configuration allows the quadrotor to efficiently transport payloads to various locations by casting, enabling more applications. However, this passive linkage introduces control inaccuracies, especially in release velocity, which are amplified during the payload's descent and lead to large casting errors.

B. Planning and Control for Suspended Payload Systems

Existing research primarily focuses on trajectory generation and control. Several previous studies [17], [18] aimed to minimize the swing angle of the suspended payload to stability. However, they often restricted the payload's range of motion, which limits its ability to execute agile maneuvers. Sreenath et al. [7] introduced a differential flatness model coupled with a geometric controller, while Yu et al. [19] proposed a nonlinear backstepping controller to enhance tracking performance. While these works represent significant progress, they primarily focus on control strategies without considering the planning aspect.

For trajectory generation, differential flatness-based methods [20], [21] and model predictive control (MPC) [8], [22], [23] are widely adopted. MPC is particularly effective for real-time obstacle avoidance [23], [24], [25] and perception-aware planning [8]. However, MPC-based planners often suffer from short prediction horizons, limiting their ability to generate long-term trajectories in complex environments. To bridge this gap, Li et al. [6] developed a differential flatness-based planner capable of real-time generation. Wang et al. [9] further advanced this by enabling mode-switching planning and control. Nevertheless, these works primarily concentrate on the transportation phase, leaving the complete delivery process unaddressed.

C. Casting and Picking

For payload casting, some studies [26], [27] formulated the planning problem as a mixed-integer quadratic program and demonstrated casting capabilities in simulation. However, their computational complexity makes them unsuitable for real-time or long-term trajectory generation. Foehn et al. [4] achieved real-world throwing tasks with maneuvers. Panetsos et al. [28] improved casting performance by incorporating target trajectory prediction into nonlinear MPC. Nevertheless, existing casting methods generally assume idealized control accuracy and neglect error amplification during payload release. As a result, ensuring safety and accuracy in real-world casting remains a challenge. Although robotic arm throwing has explicitly addressed such uncertainty via sensitivity analysis [29] or invariance control [30], [31], [32], these fully actuated solutions are not directly transferable to the underactuated dynamics of suspended payload systems.

Regarding payload picking, Luo et al. [33] achieved notable results in picking performance by utilizing a 1-DoF robotic arm. Spica et al. [34] computed a time-optimal trajectory using primitives and validated their approach in simulations. However, this work overlooks collision avoidance and only focuses on the picking stage. Antal et al. [5] considered the complete delivery process, but picking and casting positions are selected via fixed waypoints. Planning these stages separately or using fixed waypoints restricts the solution space and can lead to time-inefficient trajectories or even failure in complex environments. A key contribution of our work is a compact optimization formulation that jointly optimizes all three stages—picking, transportation, and casting—to adaptively generate safe and time-efficient trajectories.

Furthermore, integrating all three stages makes the nonlinear optimization highly sensitive to initialization. Previous approaches [6] initialize only positions, which suffices for transportation but is inadequate for casting, where an incorrect release velocity leads to infeasible trajectories or poor local minima. We address this with a hierarchical initialization strategy that provides high-quality guesses for both position and velocity.

In summary, existing approaches address only isolated aspects of the delivery process, treating picking, transportation, and casting separately, neglecting the impact of control errors on accuracy, or overlooking initialization for complex dynamic tasks. In contrast, our framework overcomes these limitations.

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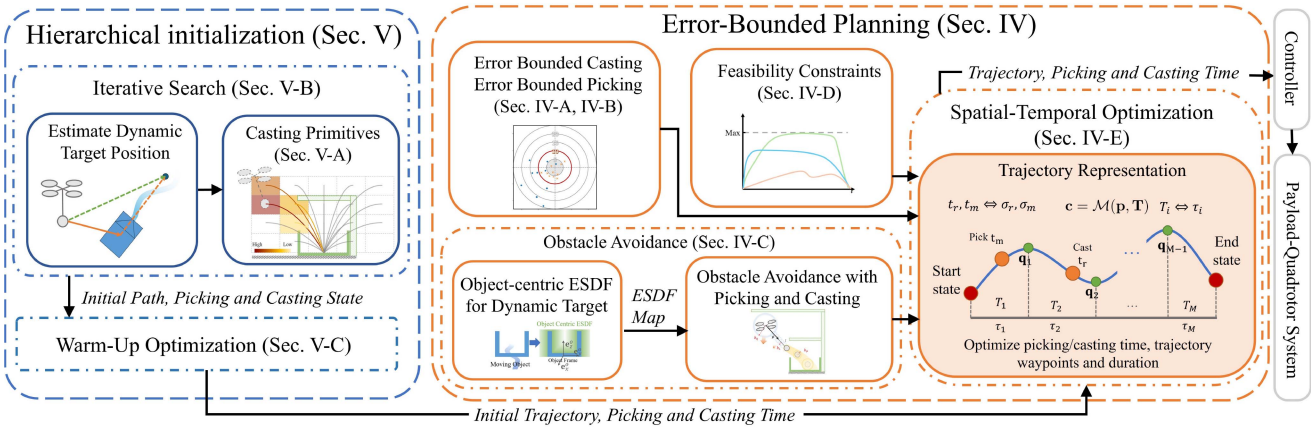


Fig. 2. Overview of the proposed system architecture. The hierarchical initialization module produces high-quality initial guesses via iterative search and warm-up optimization. The error-bounded planning module considers control errors in the payload's state and dynamic targets to bound the landing and picking error within a certain range.

III. Preliminary

Before detailing our proposed method, this section establishes the foundational elements of our work.

A. Problem Formulation

The problem considered in this article is autonomous payload delivery using a quadrotor with a cable-suspended payload in a cluttered 3-D environment. The quadrotor is tasked with picking up a payload from a platform $\mathbf{P}_o$, transporting it, and casting it into a designated area $\mathbf{P}_d$. The picking and casting targets may be moving in a two-dimensional plane during the operation. The goal is to generate a trajectory $\mathcal{Z}_i(t)$ with picking and casting to achieve in-flight payload delivery. Due to the control error in the system, the planner should consider the uncertainty ξ in the payload's state and the dynamic targets. The picking and casting error should be restricted to a certain range. The trajectory should be collision-free and dynamically feasible. The objective is to minimize the control efforts and the delivery time.

B. System Overview

The proposed system architecture is depicted in Fig. 2. The planning pipeline comprises two components: hierarchical initialization (see Section V), which produces promising initial guesses via an iterative search and a warm-up optimization, and error-bounded planning (see Section IV), which refines this solution through a joint spatial-temporal optimization that minimizes control effort and delivery time while bounding the landing or picking error under uncertainties in both the payload's state and the dynamic targets. The generated trajectory is tracked by a controller.

C. System Dynamics and Differential Flatness

The configuration of a quadrotor with a suspended payload system is shown in Fig. 3. We assume the cable remains taut in the transportation, which is an energy-efficient manner [6]. The system is approximated as a massless rod connecting two mass

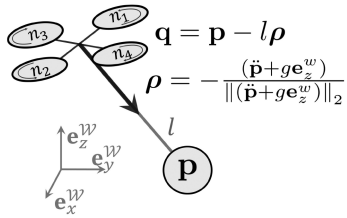


Fig. 3. Configuration of a quadrotor with a suspended payload system.

points. Following d'Alembert principle [8], we have (1) and (2)

$$m_Q l\ddot{\boldsymbol{\rho}} + m_Q l(\dot{\boldsymbol{\rho}}\cdot\dot{\boldsymbol{\rho}})\boldsymbol{\rho} = \boldsymbol{\rho}\times(\boldsymbol{\rho}\times f_t\mathbf{R}\mathbf{e}_z^w) \tag{1}$$

$$(\boldsymbol{\rho}\cdot f_t\mathbf{R}\mathbf{e}_z^w - m_Q l(\dot{\boldsymbol{\rho}}\cdot\dot{\boldsymbol{\rho}}))\,\boldsymbol{\rho} = (m_Q+m_L)\,(\ddot{\mathbf{p}}+g\mathbf{e}_z^w)\,. \tag{2}$$

$\boldsymbol{\rho}$ is a unit vector from the quadrotor to the payload center, representing the cable direction. l indicates the cable length, while $\mathbf{p}$ and $\mathbf{q}$ represent the payload's and quadrotor's positions, respectively. The variable f_t represents the quadrotor's thrust and $\mathbf{R}$ is its rotation matrix. In addition, $\mathbf{e}_z^w$ is the z-axis of the world frame.

To empower computationally efficient trajectory planning, differential flatness is utilized to reduce the state dimension and simplify the optimization problem [35]. For the quadrotor with a suspended payload, the flat outputs are chosen as $\mathcal{Z} = [\mathbf{p}, \psi]$, where $\mathbf{p}$ is the payload's position and ψ is the quadrotor's yaw angle [20]. With this formulation, all states and control inputs can be expressed in terms of these flat outputs and their derivatives. In particular, the cable direction $\boldsymbol{\rho}$ is determined by

$$\boldsymbol{\rho} = -\frac{(\ddot{\mathbf{p}}+g\mathbf{e}_z^w)}{\|(\ddot{\mathbf{p}}+g\mathbf{e}_z^w)\|_2} \tag{3}$$

$$\mathbf{q} = \mathbf{p} - l\boldsymbol{\rho}. \tag{4}$$

Thus, once the payload's trajectory $\mathbf{p}(t)$ is planned, the corresponding quadrotor position $\mathbf{q}(t)$ and its derivatives can be deduced using (4) and (3). The remaining states are derived similarly to a conventional quadrotor [36].

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IV. Error-Bounded Planning for Casting and Picking

The accuracy of payload delivery is often compromised by the precision limitations of the controller. The inherent tracking error in the controller can induce slight deviations in the payload's release position and velocity, which may amplify over time and significantly affect both the casting trajectory and landing precision of the payload. In addition, the dynamic targets further complicate the casting and picking accuracy. To bridge these gaps, we propose an error-bounded planning method for casting and picking. This method leverages robust optimization to limit the worst-case picking and casting errors for the modeled uncertainty sets. Under these assumptions, the planner constrains the delivery error and collision risk within prescribed bounds. Our method jointly optimizes the picking, transportation, and casting trajectory to achieve the seamless delivery of the payload.

A. Error-Bounded Casting in Dynamic Environment

This section models the payload's landing error and derives its bound. We employ a robust optimization method that considers the worst-case casting error to restrict the payload's landing accuracy. The payload's landing point $\tilde{\mathbf{p}}_f^{xy}$ is determined by the payload's release position $\mathbf{p}_r^{xy} = [x_r, y_r]^T$ and velocity $\mathbf{v}_r^{xy} = \dot{\mathbf{p}}_r^{xy} = [\dot{x}_r, \dot{y}_r]^T$, obtained from the trajectory. Here, the subscript "f" denotes the final landing point. For convenience, we denote all the payload states at the release moment as $\mathbf{X}_r = [\mathbf{p}_r, \dot{\mathbf{p}}_r, \dots]$. Due to the controller's error or the uncertainty from environments, the release point and velocity deviate slightly from their desired values. The actual landing point $\tilde{\mathbf{p}}_f^{xy}$ is expressed as (5) and (6)

$$\tilde{\mathbf{p}}_f^{xy} = \mathbf{p}_r^{xy} + \xi_{\mathbf{p}} + (\mathbf{v}_r^{xy} + \xi_{\mathbf{v}})\tilde{t}_f \tag{5}$$

$$\tilde{t}_f = \left(\dot{z}_r + \xi_{v_z} + \sqrt{(\dot{z}_r + \xi_{v_z})^2 + 2g(z_r - z_g + \xi_z)}\right)/g \tag{6}$$

where z_r and $\dot{z}_r$ are the release height and vertical velocity of the payload. z_g denotes the ground height. The payload flight time $\tilde{t}_f$ is the duration from the payload release to the landing instant. The deviations ξ are random variables representing the discrepancies between the desired and actual release states, written as follows:

$$\xi = \{(\xi_{\mathbf{p}}, \xi_{\mathbf{v}}, \xi_z, \xi_{v_z}) | \xi_i \in \mathcal{U}_i\}, \quad \mathcal{U}_i = \{\|\xi_i\|_2 \leq \varepsilon_i\}. \tag{7}$$

Here, $\mathcal{U}_i$ is the uncertainty set for each deviation, with $\varepsilon_i \in \mathbb{R}^+$ representing the maximum error. Specifically, $\xi_{\mathbf{p}} \in \mathbb{R}^2$ and $\xi_{\mathbf{v}} \in \mathbb{R}^2$ denote the deviations of the position and velocity on the horizontal plane, while $\xi_z \in \mathbb{R}$ and $\xi_{v_z} \in \mathbb{R}$ refer to the deviations in the height and vertical velocity. The values $\varepsilon_{\mathbf{p}}$, $\varepsilon_{\mathbf{v}}$, ε_z, and ε_{v_z} are the upper bounds of the errors in horizontal position, horizontal velocity, vertical position, and vertical velocity, respectively. Note that the control error bounds can not only be a constant, but also a function of the system states.

The desired casting target point $\tilde{\mathbf{p}}_d^{xy}$ is defined as follows:

$$\tilde{\mathbf{p}}_d^{xy} = \mathbf{p}_d(t_r + \tilde{t}_f) + \xi_{\mathbf{p}_d} \tag{8}$$

where the subscript "d" denotes the desired casting target, and $\mathbf{p}_d(*)$ is the nominal trajectory of the casting target. t_r is the time of the payload's release. $\xi_{\mathbf{p}_d} \in \mathcal{U}_{\mathbf{p}_d}$ is the deviation of the casting target. The uncertainty set for the casting target is given by the following equation:

$$\mathcal{U}_{\mathbf{p}_d} = \left\{\xi_{\mathbf{p}_d} \in \mathbb{R}^2 \,\middle|\, \|\xi_{\mathbf{p}_d}\|_2 \leq \varepsilon_{\mathbf{p}_d}\right\}. \tag{9}$$

The payload landing error $\tilde{e}(\mathbf{X}_r, t_r, \xi)$ is defined as the Euclidean distance between the actual landing point and the target point. The landing error constraint is formulated as follows:

$$\mathcal{C}_e(\mathbf{X}_r, t_r) = \sup_{\xi \in \mathcal{U}} \tilde{e}(\mathbf{X}_r, t_r, \xi) - \delta_e \leqslant 0. \tag{10}$$

The δ_e is the maximum allowable landing error. We need to find the supremum of $\tilde{e}(\mathbf{X}_r, t_r, \xi)$. Applying the Minkowski inequality, we obtain the following equation:

$$\begin{aligned}
\tilde{e}(\mathbf{X}_r, t_r, \xi) &= \left\|\tilde{\mathbf{p}}_f^{xy} - \tilde{\mathbf{p}}_d^{xy}\right\|_2 \\
&= \left\|\mathbf{p}_r^{xy} + \mathbf{v}_r^{xy}\tilde{t}_f - \mathbf{p}_d(t_r + \tilde{t}_f) + \xi_{\mathbf{p}} + \xi_{\mathbf{v}}\tilde{t}_f - \xi_{\mathbf{p}_d}\right\|_2 \\
&\leq \left\|\mathbf{p}_r^{xy} + \mathbf{v}_r^{xy}\tilde{t}_f - \mathbf{p}_d(t_r + \tilde{t}_f)\right\|_2 + \left\|\xi_{\mathbf{p}} + \xi_{\mathbf{v}}\tilde{t}_f - \xi_{\mathbf{p}_d}\right\|_2 \\
&\leq \left\|\mathbf{p}_r^{xy} + \mathbf{v}_r^{xy}\tilde{t}_f - \mathbf{p}_d(t_r + \tilde{t}_f)\right\|_2 + \varepsilon_{\mathbf{p}} + \varepsilon_{\mathbf{v}}\tilde{t}_f + \varepsilon_{\mathbf{p}_d}.
\end{aligned} \tag{11}$$

In this equation, equality holds if and only if $\mathbf{p}_r^{xy} + \mathbf{v}_r^{xy}\tilde{t}_f - \mathbf{p}_d(\tilde{t}_f + t_r)$, $\xi_{\mathbf{p}}$, $\xi_{\mathbf{v}}$, and $-\xi_{\mathbf{p}_d}$ are collinear. Note that $\tilde{t}_f$ is an uncertain variable depending on the deviations of the release height and velocity, ξ_z and ξ_{v_z}. Moreover, the nominal casting target $\mathbf{p}_d(t_r + \tilde{t}_f)$ can vary with different $\tilde{t}_f$. To simplify the problem, we linearize $\mathbf{p}_d(t_r + \tilde{t}_f)$ at the nominal landing moment $t_r + t_f$. t_f is the nominal flight time. The linearization is given by the following equation:

$$\mathbf{p}_d(t_r + \tilde{t}_f) \approx \mathbf{p}_{d|t_r+t_f} + \dot{\mathbf{p}}_{d|t_r+t_f}(\tilde{t}_f - t_f). \tag{12}$$

Substituting (12) into (11), the landing error can be approximated as follows:

$$\begin{aligned}
&\sup_{\xi \in \mathcal{U}} \tilde{e}(\mathbf{X}_r, t_r, \xi) \\
&\approx \left\|\mathbf{p}_r^{xy} - \mathbf{p}_{d|t_r+t_f} + \dot{\mathbf{p}}_{d|t_r+t_f} t_f + (\mathbf{v}_r^{xy} - \dot{\mathbf{p}}_{d|t_r+t_f})\tilde{t}_f\right\|_2 \\
&\quad + \varepsilon_{\mathbf{p}} + \varepsilon_{\mathbf{v}}\tilde{t}_f + \varepsilon_{\mathbf{p}_d}.
\end{aligned} \tag{13}$$

Now, in (13), the supremum value of $\tilde{e}(\mathbf{X}_r, t_r, \xi)$ is related with $\tilde{t}_f$. For convenience, we define $\sup_{\xi \in \mathcal{U}} \tilde{e}(\mathbf{X}_r, t_r, \xi)$ as $E(\tilde{t}_f)$. As proven in Appendix A, $E(\tilde{t}_f)$ is a convex function with respect to $\tilde{t}_f$. Therefore, its maximum value over the interval $[\min(\tilde{t}_f), \max(\tilde{t}_f)]$ must occur at one of the endpoints. Consequently, the uncertain constraint can be converted into two deterministic constraints as (14)

$$\begin{aligned}
\tilde{e}_{1,2} = &\left\|\mathbf{p}_r^{xy} - \mathbf{p}_{d|t_r+t_f} + \dot{\mathbf{p}}_{d|t_r+t_f} t_f + (\mathbf{v}_r^{xy} - \dot{\mathbf{p}}_{d|t_r+t_f}) t_f^{\pm}\right\|_2 \\
&+ \varepsilon_{\mathbf{p}} + \varepsilon_{\mathbf{v}} t_f^{\pm} + \varepsilon_{\mathbf{p}_d}
\end{aligned}$$

where $t_f^+ = \max_{\xi \in \mathcal{U}} \tilde{t}_f$ and $t_f^- = \min_{\xi \in \mathcal{U}} \tilde{t}_f$ correspond to $\tilde{e}_1$ and $\tilde{e}_2$, respectively

$$\tilde{e}_1 - \delta_e \leqslant 0, \quad \tilde{e}_2 - \delta_e \leqslant 0. \tag{14}$$

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From (6), the flight time $\tilde{t}_f$ is monotonically increasing with respect to the vertical release deviations ξ_z and ξ_{v_z}. A detailed proof is provided in Appendix B. Therefore, we have

$$\begin{aligned}\arg\max_{\xi\in\mathcal{U}}(\tilde{t}_f) &= \{\xi_z = \varepsilon_z, \xi_{v_z} = \varepsilon_{v_z}\}\\ \arg\min_{\xi\in\mathcal{U}}(\tilde{t}_f) &= \{\xi_z = -\varepsilon_z, \xi_{v_z} = -\varepsilon_{v_z}\}.\end{aligned} \tag{15}$$

We also add a soft regularization $\lambda_e(\tilde{e}_1^2 + \tilde{e}_2^2)$ on the landing error to improve robustness across environments.

B. Error-Bounded Picking in Dynamic Environment

In our system, we use an electromagnet to pick up objects. Similar to casting, the picking process also suffers from small deviations in the electromagnet's position. Moreover, when dealing with dynamic objects, there are additional uncertainties arising from imperfect observations and prediction errors. To improve picking accuracy, we need to account for these uncertainties.

1) Horizontal Pick Error Constraint: For a successful grasp, the electromagnet must be precisely aligned with the object in the horizontal plane. We define the actual horizontal pick position (i.e., the electromagnet's position, denoted by the subscript "m") as $\tilde{\mathbf{p}}_m^{xy}$ and the object's horizontal position (denoted by the subscript "o") as $\tilde{\mathbf{p}}_o^{xy}$, expressed as (16) and (17)

$$\tilde{\mathbf{p}}_m^{xy} = \mathbf{p}_m^{xy} + \xi_{\mathbf{p}} \tag{16}$$

$$\tilde{\mathbf{p}}_o^{xy} = \mathbf{p}_o(t_m) + \xi_{\mathbf{p}_o} \tag{17}$$

where $\mathbf{p}_m^{xy}$ is the nominal pick position, and $\mathbf{p}_o(t_m)$ is the nominal object's position at the pick moment. The deviations $\xi_{\mathbf{p}}$ and $\xi_{\mathbf{p}_o}$ represent the uncertainty of the horizontal pick position and the object position, respectively. In particular, $\xi_{\mathbf{p}_o}$ denotes horizontal position deviation of the object, which is defined by the uncertainty set in the following equation:

$$\xi_{\mathbf{p}_o} \in \mathcal{U}_{\mathbf{p}_o} = \left\{\xi_{\mathbf{p}_o} \in \mathbb{R}^2 \,\middle|\, \|\xi_{\mathbf{p}_o}\|_2 \leqslant \varepsilon_{\mathbf{p}_o}\right\}. \tag{18}$$

The horizontal picking error is defined as the Euclidean distance between the pick position and the object's position. The horizontal picking error is formulated as follows:

$$\begin{aligned}\tilde{e}_m(\mathbf{X}_m, t_m, \xi) &= \|\tilde{\mathbf{p}}_m^{xy} - \tilde{\mathbf{p}}_o^{xy}\|_2\\ &= \|\mathbf{p}_m^{xy} + \xi_{\mathbf{p}} - \mathbf{p}_o(t_m) - \xi_{\mathbf{p}_o}\|_2\\ &\leq \|\mathbf{p}_m^{xy} - \mathbf{p}_o(t_m)\|_2 + \varepsilon_{\mathbf{p}} + \varepsilon_{\mathbf{p}_o}\end{aligned} \tag{19}$$

where $\mathbf{X}_m$ is the system state at the pick moment. The worst-case (supremum) of $\tilde{e}_m(\mathbf{X}_m, t_m, \xi)$ is attained when the directions of $\mathbf{p}_m^{xy} - \mathbf{p}_o(t_m)$, $\xi_{\mathbf{p}}$, and $-\xi_{\mathbf{p}_o}$ are aligned. The horizontal pick error constraint is written as follows:

$$\mathcal{C}_m(\mathbf{X}_m, t_m) = \sup_{\xi\in\mathcal{U}} \tilde{e}_m(\mathbf{X}_m, t_m, \xi) - \delta_m \leqslant 0. \tag{20}$$

where δ_m is the maximum allowable pick error. We similarly add a regularization term $\lambda_m \tilde{e}_m^2$ to penalize the horizontal pick error.

2) Vertical Pick Error Constraint: Using an electromagnet to pick up objects requires precise vertical control to ensure contact with the payload. The vertical pick position $\tilde{Z}_m$ is written as follows:

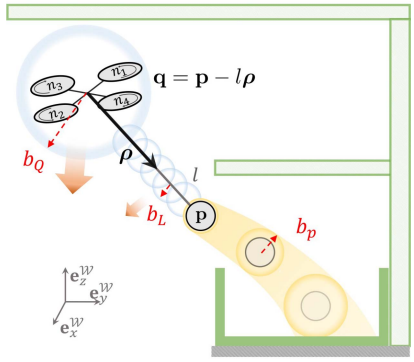


Fig. 4. Casting the payload into a box beneath a shelf. The quadrotor-payload system is simplified as a bunch of bubbles. To avoid collision with the shelf, the payload must maintain a safe distance b_p from the shelf. The safe distance b_p is calculated from the max error range.

$$\tilde{Z}_m = z_m + \xi_z \tag{21}$$

where z_m is the nominal pick height.

Similarly, we assume the object's height is z_o, with deviation $\xi_{z_o} \in \mathcal{U}_{z_o}$, defined as follows:

$$\tilde{Z}_o = z_o + \xi_{z_o}, \quad \mathcal{U}_{z_o} = \left\{\xi_{z_o} \in \mathbb{R} \,\middle|\, |\xi_{z_o}| \leq \varepsilon_{z_o}\right\}. \tag{22}$$

The vertical pick error can be written as follows:

$$\tilde{e}_z(\mathbf{X}_m, \xi) = \tilde{Z}_m - \tilde{Z}_o = z_m + \xi_z - z_o - \xi_{z_o}. \tag{23}$$

To guarantee that the electromagnet makes contact with the object, the vertical pick error should be nonpositive. Thus, the vertical pick error constraint is formulated as follows:

$$\mathcal{C}_z(\mathbf{X}_m) = \sup_{\xi\in\mathcal{U}} \tilde{e}_z(\mathbf{X}_m, \xi) = z_m - z_o + \varepsilon_z + \varepsilon_{z_o} \leqslant 0. \tag{24}$$

C. Collision Avoidance

Safely delivering payloads in complex environments involves two essential requirements: ensuring that the payload's projectile does not collide with obstacles and ensuring the entire system can navigate through narrow spaces safely. Also, considering the dynamic targets, we need to ensure safe picking and casting with dynamic targets.

1) Safety in Casting: The first requirement focuses on the payload's casting. When payloads must be delivered in a narrow space, we need to ensure that the payload trajectory does not collide with obstacles. The casting trajectory should pass through the narrow space safely, like Fig. 4. Here, we propose an error-bounded collision avoidance constraint for the payload's projectile.

After the payload is released, its position $\tilde{\mathbf{p}}_p$ at time t_p is

$$\tilde{\mathbf{p}}_p(t_p) = \mathbf{p}_r + \xi_{\mathbf{p}_r} + (\mathbf{v}_r + \xi_{\mathbf{v}_r})\, t_p + \frac{1}{2} g t_p^2 \mathbf{e}_z^w \tag{25}$$

$$\xi_{\mathbf{p}_r} = \left[\xi_{\mathbf{p}}^T, \xi_z\right]^T, \quad \xi_{\mathbf{v}_r} = \left[\xi_{\mathbf{v}}^T, \xi_{v_z}\right]^T \tag{26}$$

where we denote the nominal release position and velocity of the payload, while $\xi_{\mathbf{p}_r}$ and $\xi_{\mathbf{v}_r}$ represent their corresponding

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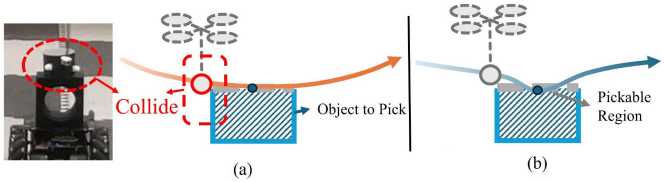


Fig. 5. Object-centric ESDF for safe interaction. (a) Without awareness of the object's geometry, the planner generates the orange path, which is more energy-efficient but results in a collision. (b) Planner uses the object-centric ESDF to treat the object's other sides as an obstacle, successfully generating a safe and feasible trajectory (blue) to access the picking region.

deviations. In order to avoid collision, the payload must keep a safe distance b_p from obstacles, defined as follows:

$$\begin{aligned} b_p(t_p) &= \sup_{\xi \in \mathcal{U}} \|\tilde{\mathbf{p}}_p(t_p) - \mathbb{E}_\xi[\tilde{\mathbf{p}}_p(t_p)]\|_2 \\ &= \sup_{\xi \in \mathcal{U}} \|\xi_{\mathbf{p}_r} + \xi_{\mathbf{v}_r} t_p\|_2 \\ &= \sqrt{\|\varepsilon_{\mathbf{p}} + \varepsilon_{\mathbf{v}} t_p\|_2^2 + (\varepsilon_z + \varepsilon_{v_z} t_p)^2} \end{aligned} \tag{27}$$

where the uncertainties $\varepsilon_{\mathbf{p}}$ and $\varepsilon_{\mathbf{v}}$ are the maximum deviations in horizontal position and velocity, respectively. The ε_z and ε_{v_z} are the maximum deviations in height and vertical velocity.

We maintain a Euclidean signed distance field (ESDF) $\mathcal{M}$ for querying the shortest distance to obstacles [37]. When the payload violates the safe distance requirement, the gradient of the ESDF will exert a repulsive force to push it away from the obstacles. The collision avoidance constraint is formulated as follows:

$$\mathcal{C}_p(\mathbf{X}_r) = b_p(t_p) + b_L - \mathcal{M}(\mathbb{E}_\xi[\tilde{\mathbf{p}}_p(t_p)]) \leqslant 0 \tag{28}$$

where b_L is the safe distance for the payload and $\mathbf{X}_r$ denotes the relevant system state at release.

2) Object-Centric ESDF: Safe interaction with dynamic targets requires avoiding collisions not only with the static environment but also with the physical volume of the moving target itself. A critical application is during the picking phase, where the planner must generate a trajectory that navigates around the moving object's body to safely approach the payload. A planner unaware of the object's geometry might generate a physically impossible path, as illustrated in Fig. 5. Thus, the object geometry must be considered during trajectory generation.

While simpler approximations, such as bounding spheres and cylinders, are commonly used for obstacle avoidance of dynamic objects, they are insufficient for close-proximity interaction tasks with complex shapes. A bounding sphere large enough to enclose the object (e.g., a truck bed) inflates the obstacle and often blocks the feasible path to the picking patch. A smaller sphere misses corners or edges and allows unsafe approaches. We, therefore, adopt an object-centric ESDF (OESDF), which provides accurate signed distances to the true object geometry while maintaining computational efficiency.

A conventional world-frame ESDF must be recomputed whenever the object moves, which is computationally expensive. To avoid this cost, we precompute an ESDF map $\mathcal{M}_o$ once in the object's coordinate frame, as shown in Fig. 6. During optimization, a point ${}^w\mathbf{p}$ (e.g., the quadrotor or payload) in the world frame is transformed into the object's frame via (29)

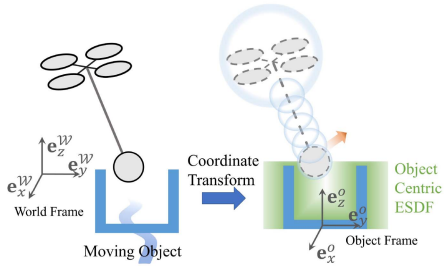


Fig. 6. Object-centric ESDF. To efficiently check for collisions with a moving object, we compute its distance field once in its local frame (right). During planning, the quadrotor's and payload's position is transformed from the world frame (left) into the object's frame to perform distance and gradient queries. Then, the distance gradient is transformed back to the world frame.

$$ {}^o\mathbf{p} = {}^w\mathbf{R}_o^T \left({}^w\mathbf{p} - {}^w\mathbf{p}_o\right) \tag{29}$$

where ${}^w\mathbf{p}_o$ and ${}^w\mathbf{R}_o$ denote the object's position and rotation matrix in the world frame, respectively.

The signed distance is then queried directly from the static map $\mathcal{M}_o({}^o\mathbf{p})$. Finally, we transform the distance gradient back to the world frame by the following equation:

$$\nabla\mathcal{M}_w({}^w\mathbf{p}) = {}^w\mathbf{R}_o \nabla\mathcal{M}_o({}^o\mathbf{p}). \tag{30}$$

In practice, to ensure the planner can always find a valid path, we set a collision-free region around the designated picking and casting regions on the object's surface. For example, for a box-shaped obstacle, we treat the top surface as collision-free, and the other sides are obstacles. This approach allows us to generate the ESDF only once in the object's frame, thereby saving computation time.

3) Whole-Body Safety: To ensure safe navigation through complex environments, it is essential to prevent collisions for the entire quadrotor-payload system, whose shape changes dynamically during flight. For collision avoidance of the whole-body, we adopt an approach similar to a previous work [6]. For a quadrotor with a suspended payload, the system's shape varies with its acceleration. Exploiting the differential flatness property, the system shape can be determined by the payload's position $\mathbf{p}$ and acceleration $\ddot{\mathbf{p}}$. We approximate this dynamically varying shape by a collection of spherical bubbles, as shown in Fig. 4. We enforce that the distance to obstacles is always greater than a specified safe distance for both the quadrotor and the payload, denoted by b_Q and b_L, respectively. This enables the system to navigate through narrow spaces safely.

D. Feasibility Constraints

Dynamic feasibility is essential for trajectory execution. We leverage differential flatness to impose dynamic constraints on the flat output space.

1) Payload Landing Velocity Constraints: Excessive landing velocity can potentially cause damage to the payload. We restrict the landing velocity to a feasible range. Also, keeping the landing velocity similar to the target's velocity can improve the success rate of the casting. This is because a lower relative velocity between the payload and the target reduces the sensitivity of the final landing error to uncertainties in the flight time. We

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divide the landing velocity into two parts: the horizontal and the vertical velocity. Considering the uncertainty, the horizontal landing velocity $\tilde{\mathbf{v}}_f^{xy}$ and the moving target velocity $\tilde{\mathbf{v}}_d^{xy}$ are written as follows:

$$\tilde{\mathbf{v}}_f^{xy} = \mathbf{v}_r^{xy} + \xi_{\mathbf{v}}, \quad \tilde{\mathbf{v}}_d^{xy} = \mathbf{v}_d^{xy} + \xi_{\mathbf{v}_d} \tag{31}$$

where $\mathbf{v}_r^{xy}$ and $\xi_{\mathbf{v}}$ are the nominal payload's horizontal release velocity and its deviation, and $\mathbf{v}_d^{xy}$ with $\xi_{\mathbf{v}_d}$ are the nominal target's horizontal velocity and its deviation, respectively. Here, we define $\xi_{\mathbf{v}} \in \mathcal{U}_{\mathbf{v}} = \{\xi_{\mathbf{v}} \in \mathbb{R}^2 \mid \|\xi_{\mathbf{v}}\|_2 \leq \varepsilon_{\mathbf{v}}\}$ and $\xi_{\mathbf{v}_d} \in \mathcal{U}_{\mathbf{v}_d} = \{\xi_{\mathbf{v}_d} \in \mathbb{R}^2 \mid \|\xi_{\mathbf{v}_d}\|_2 \leq \varepsilon_{\mathbf{v}_d}\}$. The upper bounds of the uncertainty are $\varepsilon_{\mathbf{v}}$ and $\varepsilon_{\mathbf{v}_d}$. The horizontal landing velocity constraint is then formulated as follows:

$$\begin{aligned} \mathcal{C}_{fv}(\mathbf{X}_r) &= \|\tilde{\mathbf{v}}_f^{xy} - \tilde{\mathbf{v}}_d^{xy}\|_2 \\ &\leq \|\mathbf{v}_r^{xy} - \mathbf{v}_d^{xy}\|_2 + \|\xi_{\mathbf{v}} - \xi_{\mathbf{v}_d}\|_2 \\ &\leq \|\mathbf{v}_r^{xy} - \mathbf{v}_d^{xy}\|_2 + \varepsilon_{\mathbf{v}} + \varepsilon_{\mathbf{v}_d} \leq v_{f_{\max}} \end{aligned} \tag{32}$$

where $v_{f_{\max}}$ is the maximum allowable horizontal landing velocity.

The vertical landing velocity is given by the following equation:

$$\tilde{\mathbf{v}}_f^{z} = -\sqrt{(\dot{z}_r + \xi_{v_z})^2 + 2g(z_r - z_g + \xi_z)}. \tag{33}$$

We assume the target's vertical velocity is zero at the landing moment. The vertical landing velocity constraint is imposed as $\|\tilde{\mathbf{v}}_f^z\|_2^2 \leqslant v_{z_{\max}}^2$. $v_{z_{\max}}$ is the maximal vertical landing velocity. From (33), the deterministic form of the vertical constraint is defined as follows:

$$\mathcal{C}_{fz}(\mathbf{X}_r) = \left(\dot{z}_r \pm \varepsilon_{v_z}\right)^2 + 2g(z_r - z_g + \varepsilon_z) \leq v_{z_{\max}}^2. \tag{34}$$

2) Payload Picking Velocity Soft Constraint: To improve the picking success rate, the electromagnet's velocity $\mathbf{v}_m$ should approximate the object's velocity $\mathbf{v}_o$. Since a hard constraint may cause infeasibility, we enforce this softly via a regularization term $\lambda_{v_m}\|\mathbf{v}_m - \mathbf{v}_o\|_2^2$, which drives the relative velocity toward zero to reduce impact shock while preserving optimization flexibility.

3) Picking Angle Constraints: The electromagnet should pick up the object at an appropriate angle to ensure successful contact. Therefore, we impose an angle range constraint at the picking point. The constraint is formulated as follows:

$$\mathcal{C}_{\beta}(\mathbf{X}_m) = \cos(\beta_{\max}) + \mathbf{e}_z^{wT}\rho \leqslant 0 \tag{35}$$

where $\beta_{\max}$ is the maximal allowable picking angle and ρ is the cable or relative direction vector.

4) Quadrotor Dynamic Constraints: We limit the velocity and acceleration of the quadrotor. These two values are derived from the payload's states by differential flatness mentioned in Section III-C. The constraints are written as (36) and (37)

$$\mathcal{C}_{Qv}(\mathbf{X}) = \|\dot{\mathbf{p}} - l\dot{\boldsymbol{\rho}}\|_2^2 - v_{Q\max}^2 \leqslant 0 \tag{36}$$

$$\mathcal{C}_{Qa}(\mathbf{X}) = \|\ddot{\mathbf{p}} - l\ddot{\boldsymbol{\rho}}\|_2^2 - a_{Q\max}^2 \leqslant 0 \tag{37}$$

where $v_{Q\max}$ and $a_{Q\max}$ represent the maximum allowable velocity and acceleration of the quadrotor, respectively.

5) Payload Dynamic Constraint: We impose the payload's velocity and acceleration constraints to ensure the payload can be transported safely, which is similar to the quadrotor's dynamic constraints. In addition, to prevent the payload from swinging excessively, we limit its maximum swing angle. This constraint is written as follows

$$\mathcal{C}_{\gamma}(\mathbf{X}) = \cos(\gamma_{\max}) + \mathbf{e}_z^{wT}\rho \leqslant 0 \tag{38}$$

where $\gamma_{\max}$ is the maximum allowable swing angle.

6) Cable Tension Constraint: In our dynamic model, we assume that the cable is always under tension. To maintain this assumption, we impose a cable tension constraint. The constraint is simplified as follows:

$$\mathcal{C}_{f_r}(\mathbf{X}) = \epsilon - g - \ddot{\mathbf{x}}_L\mathbf{e}_z^{w} \leqslant 0 \tag{39}$$

where ϵ is a small positive constant that enforces a margin in the inequality. This ensures the payload's vertical acceleration remains strictly less than gravity, maintaining a positive tension in the cable.

E. Spatial-Temporal Optimization

We formulate a unified nonlinear optimization that simultaneously determines the trajectory and the pickup and release times, ensuring all constraints are satisfied while improving time efficiency. The trajectory is parameterized with M pieces polynomial, where each segment is $\mathcal{Z}_i(t) = \mathbf{c}_i^T\boldsymbol{\beta}(t)$, $\boldsymbol{\beta}(t) = [1, t, \ldots, t^N]^T$. Here, $\mathbf{c} = [\mathbf{c}_1, \mathbf{c}_2, \ldots, \mathbf{c}_M]$ and $\mathbf{T} = [T_1, T_2, \ldots, T_M]$ are coefficients and time allocations of each piece. We use the system's flat output, payload position $\mathbf{p}$, to represent trajectory. In our case, $\mathbf{p}$ is three-dimensional $(D = 3)$ and is modeled by a seventh-order $(N = 7)$ polynomial. A seventh-order polynomial enables minimum-snap trajectory optimization and ensures the smoothness of quadrotor acceleration [6], [9], [35]. Instead of directly using polynomial splines, we employ MINCO [35], which reformulates the polynomial coefficients $\mathbf{c}$ into intermediate waypoints $\mathbf{p}_k$ and time allocations $\mathbf{T}$ via a differentiable transformation, $\mathbf{c} = \mathcal{M}(\mathbf{p}_k, \mathbf{T})$. This method produces a trajectory that minimizes control effort while ensuring the path passes through all specified waypoints $\mathbf{p}_k$ in the predetermined times $\mathbf{T}$. The resulting trajectory inherently satisfies continuity constraints up to the required order.

The problem is formalized as follows:

$$\begin{aligned} \min_{\mathbf{p_k},\boldsymbol{\sigma},\boldsymbol{\tau}} J(\mathbf{p_k}, \boldsymbol{\sigma}, \boldsymbol{\tau}) &= \sum_{i=1}^{M}\int_0^{T_i} \mathcal{Z}_i^{(s)}(t)^T Q_J \mathcal{Z}_i^{(s)}(t)dt \\ &\quad + \lambda_T \sum_{i=1}^{M} T_i \end{aligned} \tag{40a}$$

$$\text{s.t. } \mathcal{Z}_i(t) = \mathcal{M}(\mathbf{p}_i, T_i)^T\boldsymbol{\beta}(t) \tag{40b}$$

$$\mathcal{Z}_1^{[s-1]}(0) = \mathbf{b}_0, \ \mathcal{Z}_M^{[s-1]}(t_M) = \mathbf{b}_f \tag{40c}$$

$$T_i = \mathcal{T}(\tau_i) \tag{40d}$$

$$\mathbf{X} = \mathcal{F}\left(\mathcal{Z}_i(t), \mathcal{Z}_i^{(1)}(t), \ldots, \mathcal{Z}_i^{(2s-1)}(t)\right) \tag{40e}$$

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$$t_r = \mathcal{D}(\sigma_r)\sum_{i=1}^{M} T_i, \quad t_m = \mathcal{D}(\sigma_m)\sum_{i=1}^{M} T_i \tag{40f}$$

$$t_r - t_m \geq \epsilon_m \tag{40g}$$

$$\mathbf{X}_r = [\mathcal{Z}(t_r), \mathcal{Z}^{(1)}(t_r), \ldots, \mathcal{Z}^{(2s-1)}(t_r)]^T \tag{40h}$$

$$\mathbf{X}_m = [\mathcal{Z}(t_m), \mathcal{Z}^{(1)}(t_m), \ldots, \mathcal{Z}^{(2s-1)}(t_m)]^T \tag{40i}$$

$$\mathcal{C}(\mathbf{X}, \mathbf{X}_r, \mathbf{X}_m) \leqslant 0. \tag{40j}$$

In the objective function (40a), the first term minimizes a high-order derivative of the trajectory (e.g., snap, when $s = 4$). Minimizing snap [35], [36] is a widely used objective for generating smooth trajectories that reduce control effort. Instead of limiting the total time of the trajectory, we penalize the total time in the objective function as previous works [38], [39], [40]. Treating the duration as a regularization term allows our joint spatial-temporal optimization to find a balanced solution. The boundary constraints in (40c) enforce that the trajectory meets the initial conditions $\mathbf{b}_0$ and final conditions $\mathbf{b}_f$ up to the order $s-1$. The flatness function $\mathcal{F}$ is used to derive the system's states $\mathbf{X}$. $\mathbf{X}_r$ and $\mathbf{X}_m$ define the states of the system at the release and pick instants, including position, velocity, and higher order derivatives. $\mathcal{C}$ represents the constraints derived in Sections IV-A–D, including the error-bounded accuracy constraints, collision avoidance, and dynamic feasibility limits.

The release time t_r and pick time t_m along the trajectory are determined by virtual percentages σ_r and σ_m. The t_r and t_m must be in the range from 0 to the total time of the trajectory. Therefore, the virtual percentages $\sigma_r, \sigma_m \in \mathbb{R}$ are mapped to the unit interval (0,1) using the diffeomorphic mapping $\mathcal{D}$

$$\mathcal{D}(\sigma) = \frac{0.5\sigma}{\sqrt{\sigma^2 + 100}} + 0.5. \tag{41}$$

This transformation, (41), effectively removes the boundary constraint on the percentage. In addition, the picking time must be before the release time, as illustrated in (40g). ϵ_m is the minimum time interval between the picking and release for execution.

Similarly, to guarantee that each segment duration T_i remains positive while using an unconstrained optimizer (e.g., L-BFGS), we introduce an unconstrained virtual time vector $\tau = [\tau_1, \ldots, \tau_M]$. A diffeomorphic time mapping function $\mathcal{T}$ is utilized to transform $\tau_i \in \mathbb{R}$ into a positive real duration $T_i \in \mathbb{R}^+$. We specifically choose this mapping, instead of something simpler like the exponential, to prevent large gradients to improve the numerical stability in the optimization. The mapping is based on the Taylor expansion of the exponential function, and ensures the transformation is twice continuously differentiable, written as follows:

$$T_i = \mathcal{T}(\tau_i) = \begin{cases} 1 + \tau_i + \frac{\tau_i^2}{2}, & \tau_i \geqslant 0 \\ \frac{2}{2 - 2\tau_i + \tau_i^2}, & \tau_i < 0. \end{cases} \tag{42}$$

To ensure feasibility, constraints, such as collision avoidance and dynamic limits, must be satisfied continuously along the entire trajectory. In other words, these constraints should be satisfied at every point in time along the polynomial trajectory. This continuous-time requirement leads to an infinite number of constraints. To handle these constraints in a tractable manner, we integrate the constraint violations over each trajectory segment. [35], [41]. The infinite constraints are condensed into a finite sum. Finally, we apply a smoothed L1 penalty function [42] to transform the trajectory optimization problem into an unconstrained problem. This unconstrained optimization problem is solved using the L-BFGS algorithm [43].

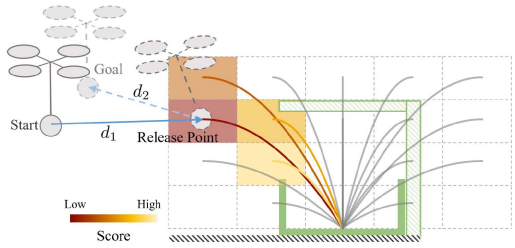


Fig. 7. Generate the casting primitives and select the best release point.

V. Hierarchical Initialization for Optimization

The trajectory optimization is hindered by complex nonlinear constraints. Therefore, a well-chosen initial guess is crucial to avoid suboptimal local minima and accelerate the convergence of the optimization process. This section presents a hierarchical initialization strategy specifically designed to generate a high-quality initial guess, thereby enabling efficient and robust convergence of the optimization.

A. Casting Primitive Generation and Evaluation

An appropriate initial release state is pivotal for casting the payload. We select the initial release state by evaluating a set of casting motion primitives. We generate a library of casting motion primitives according to the release point's position to cover a region, as delineated in Fig. 7. Specifically, we uniformly divide the region into 3-D grids. Each grid's centroid is designated as the release position $\mathbf{p}_{r_i} = [x_{r_i}, y_{r_i}, z_{r_i}]^T$. We assume the initial vertical velocity is zero, i.e., $v_{r_i}^z = 0$. The flying time t_{f_i} and horizontal release velocity $\mathbf{v}_{r_i}^{xy}$ of the payload are calculated as follows:

$$t_{f_i} = \sqrt{\frac{2(z_{r_i} - z_g)}{g}}, \quad \mathbf{v}_{r_i}^{xy} = \frac{\mathbf{p}_d^{xy} - \mathbf{p}_{r_i}^{xy}}{t_{f_i}} \tag{43}$$

where $\mathbf{p}_d^{xy}$ is the desired landing point. If the velocity $\|\mathbf{v}_{r_i}^{xy}\|_2$ exceeds the payload's maximum velocity, this primitive is deemed infeasible and thus excluded. The blank grids in Fig. 7 represent the excluded primitives. In practice, we generate the motion primitives offline with the desired landing point initially set to the origin. During online planning, the release point can be dynamically shifted to a specified position according to the actual desired landing point.

We evaluate the generated casting primitives to select the most suitable initial release point using multiple criteria. First, the safety of each primitive is checked. If the system collides at the release point, or the payload's projectile collides, the primitive is discarded. Primitives designated in gray in Fig. 7 are discarded due to safety violations.

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Subsequent to the safety assessment, each primitive is scored based on a defined metric, formalized as (44). The best release point is the one with the lowest score.

$$s_i = \alpha_{d_1} d_{1_i} + \alpha_{d_2} d_{1_i} + \alpha_v \|\mathbf{v}_{r_i}\|_2 + \alpha_t t_{f_i} \tag{44}$$

where d_{1_i} and d_{2_i} denote the distances from the start to the release point and from the release point to the goal, respectively. $\mathbf{v}_{r_i}$ is the release velocity, and t_{f_i} is the payload's flight time. $\alpha_{d_1}, \alpha_{d_2}, \alpha_v, \alpha_t$ are the weights for each term. In general, the weight α_{d_1} is set marginally higher than α_{d_2} to break ties, particularly in symmetric cases. Releasing the payload earlier is also presumed to be more energy-efficient. The terms weighted by α_v and α_t represent a preference for a lower release velocity and a shorter payload flight time, which contribute to improved landing precision.

B. Iterative Kinodynamic A* Search

Considering a dynamic picking and casting target, the precise picking or casting position depends on the interception time. The interception time depends on when the quadrotor arrives, while the arrival time is determined by the duration of the planned trajectory. But searching for a trajectory requires a target goal (picking or casting position). To resolve this circular dependency, we propose an iterative search strategy. The process begins with an initial time estimate based on the target's kinematics by Algorithm 2. This time is used to query the target's trajectory to establish an initial goal state. Subsequently, we employ an iterative refinement process: in each iteration k, the Kinodynamic A* search generates a trajectory to reach the current goal $\mathbf{g}_k$. The actual duration of this trajectory, denoted as t, is then used to update the predicted target state, which yields a refined goal $\mathbf{g}_{k+1}$ for the next iteration. This cycle repeats until the goal state converges, ensuring consistency between the planning goal and the target's motion.

The algorithm is shown in algorithm 1. $\mathbf{g}_k$ denotes the target goal state in the kth iteration. For a picking task, the goal $\mathbf{g}_k$ is directly sampled from the target's trajectory $\mathbf{P}(t)$. For a casting task, we obtained the desired casting target from $\mathbf{P}(t)$. Then, we evaluate the casting primitives based on the desired casting target and select the best release point as the goal $\mathbf{g}_k$.

1) Estimate Dynamic Target Position: Since the position of a dynamic target is time-dependent, we must first estimate the time required for the quadrotor to intercept it. We utilize this time to query the target's position as an initial path search goal. We propose an iterative algorithm to estimate this initial time. Given a candidate quadrotor speed bound v, we estimate the earliest interception time t by solving (45); Algorithm 2 increases v until a feasible t is found and uses it as the initial guess for Algorithm 1. If the maximum velocity is reached and the time is still not feasible, the algorithm reports that the problem is infeasible.

To estimate the time required to reach the target, we solve a simplified optimization problem as shown in Fig. 8. We linearize the target's trajectory at the starting point, in other words, assuming it moves at a constant velocity. We also assume the quadrotor's velocity is constant and seek to find the minimum

Algorithm 1: Iterative Kinodynamic A* Search.

Input: Current quadrotor state $\mathbf{X}_{start}$, Dynamic target trajectory $\mathbf{P}(t)$
Output: Feasible path $path$, Time to target t
$t_0 \leftarrow$ Estimate initial interception time using Algorithm 2
Get goal state $\mathbf{g}_0$ based on time t_0 from $\mathbf{P}(t_0)$.
$k \leftarrow 0$
repeat
  $path, t \leftarrow$ Kinodynamic A* Search for $\mathbf{g}_k$
  Get goal state $\mathbf{g}_{k+1}$ based on time t from $\mathbf{P}(t)$.
  $\Delta \leftarrow \|\mathbf{g}_{k+1} - \mathbf{g}_k\|$
  $k \leftarrow k + 1$
until $\Delta < \epsilon$ or $k >$ maximum iteration
return $path, t$

Algorithm 2: Estimate Initial Time.

Input: Maximum velocity v_{max}, Current quadrotor state $\mathbf{X}_{start}$, Dynamic target trajectory $\mathbf{P}(t)$
Output: Initial interception time (Time to target) t
$v \leftarrow \frac{v_{max}}{2}$ ▷ Initial speed
while $v < v_{max}$ **do**
  Compute t using equation (46) with speed bound v
  if t is feasible **then**
    return t
  end if
  $v \leftarrow v + v_{step}$
end while
return Not feasible

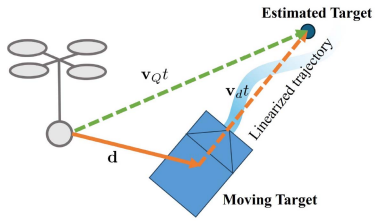


Fig. 8. Estimate the coarse time required for the quadrotor to reach the target.

time to intercept the target. The optimization problem is formulated as follows:

$$\begin{aligned} \min_{t} \quad & t \\ \text{s.t.} \quad & \mathbf{v}_Q t = \mathbf{v}_d t + \mathbf{d} \\ & \|\mathbf{v}_Q\|_2 \le v, \quad t \ge 0 \end{aligned} \tag{45}$$

where $\mathbf{v}_Q$ represents the quadrotor's velocity, $\mathbf{v}_d$ is the target's velocity, $\mathbf{d}$ is the displacement vector between the start position and the target current position. This can be calculated by the quadrotor's current state and the target's trajectory. v is the maximum allowed speed. The closed-form solution of the

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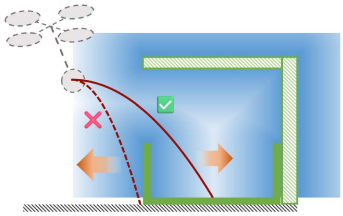


Fig. 9. Inexact initialization of position and velocity (dashed red line) traps optimization in improper local minima. A proper initialization (solid red line) from our warm-up strategy improves computational efficiency. The blue gradient represents the ESDF for collision avoidance, with arrows indicating the gradient direction away from obstacles.

optimization problem is given in the following equation:

$$t = \begin{cases} 0, & \|\mathbf{d}\|_2^2 = 0 \\ \max\{0, \frac{-\|\mathbf{d}\|_2^2}{2b}\}, & a = 0, \\ \frac{b+\sqrt{b^2-a\|\mathbf{d}\|_2^2}}{-a}, & a < 0 \text{ or } \left(0 < a \leq \frac{b^2}{\|\mathbf{d}\|_2^2} \text{ and } b < 0\right) \\ \text{infeasible}, & \text{otherwise} \end{cases} \tag{46}$$

$$a = \|\mathbf{v}_d\|_2^2 - v^2, \quad b = \mathbf{d} \cdot \mathbf{v}_d.$$

2) Kinodynamic A Search:* After determining an initial target position, we need to generate a dynamically feasible and collision-free path to it, which will serve as a high-quality skeleton for the subsequent optimization. To address this, we employ the Kinodynamic A$*$ search algorithm. The objective of path searching is to minimize both the temporal duration and the control efforts. The control input is defined as the payload acceleration $\ddot{\mathbf{x}}_L$, and the trajectories are represented by second-order time-parameterized polynomial primitives. Unlike A$*$, kinodynamic A$*$ incorporates both obstacle avoidance and dynamic constraints to generate feasible trajectories. Furthermore, it can obtain the duration of each trajectory segment, which can be utilized as initial values for the optimization problem. For obstacle avoidance, the shape of the system is simplified as a rod with a sphere at the end for collision checking. We leverage differential flatness to compute the pose of the rod using (3) and (4).

C. Warm-Up Optimization

The payload's casting trajectory is determined by its release position and velocity. Accurately initializing the payload's casting trajectory is vital for nonlinear optimization. As shown in Fig. 9, the dashed casting trajectory is trapped in a local minimum due to a poor initialization of the release velocity. With such an initialization, the optimization may fail to converge. However, it is difficult to directly initialize the release velocity of the payload because the velocity is implicitly determined by the polynomial trajectory parameterization.

To address this issue, we adopt a warm-up optimization strategy. First, feasible picking and casting states are obtained from the iterative kinodynamic search. In the warm-up optimization, we solve a simplified optimization problem. We freeze the casting and picking states, i.e., the picking and release positions and velocities are fixed to the values returned by the search, and we only optimize the transportation trajectory. We keep only the core physical feasibility constraints, including: collision avoidance, quadrotor and payload dynamic constraints, and cable tension constraint. All other constraints related to error bounds are ignored in this phase. This warm-up strategy optimizes the trajectory in a reduced constraint space, guiding the solution toward a promising region of the decision space. The primary goal of this warm-up is to steer the optimization away from poor local minima, not to find a high-quality trajectory. Therefore, we can relax the tolerance settings for this warm-up phase to improve overall computational efficiency. Finally, a joint optimization, which considers all constraints, is performed to refine all three stages of the trajectory: picking, transportation, and casting.

VI. Implementation Details

A. Control Error Bound Estimation

Accurate control error bounds are essential for reliable casting error estimation. We establish these bounds using empirical data from our quadrotor-payload system. To do this, we commanded the system to follow an eight-shaped trajectory for 12 rounds and recorded the deviations between the reference and actual payload states. This process allowed us to characterize the distributions of position and velocity tracking errors.

In our implementation, control error bounds are determined from a specified confidence level of the collected error data. We adopt percentile-based bounds, which are distribution-agnostic and directly tied to measured performance. The 98th percentile is selected as a practical high-confidence bound that avoids excessive conservatism from rare outliers while maintaining reliable performance. Note that this value can be adjusted according to application's risk tolerance.

Our framework supports two types of control error bounds: constant bounds, which use fixed values, and state-dependent bounds, which vary with the system's state. To select the control-error bound model, we analyzed the correlation between each error channel and the payload acceleration magnitude using Pearson's coefficient r, Spearman's coefficient ρ, and two-sided permutation tests with 10 000 permutations. The horizontal position error exhibits the largest correlation, with $r = 0.51$ ($p < 0.0001$, 95% confidence intervals (CI) [0.47,0.58]) and $\rho = 0.52$ ($p < 0.0001$, 95% CI [0.49,0.58]). The horizontal velocity and vertical velocity errors also show statistically supported nonzero correlations ($r = 0.33, \rho = 0.40$, and $r = 0.47, \rho = 0.43$, respectively; $p < 0.0001$), while the vertical position error does not show a statistically supported correlation ($r = 0.003$, $p = 0.7908$; $\rho = 0.06$, $p = 0.3996$). To balance model fidelity and computational efficiency, we use an acceleration-dependent second-order polynomial bound for the horizontal position error and 98th-percentile constant bounds for the remaining error channels.

B. Nonlinear Model Predictive Controller (NMPC)

The control framework employs an NMPC [6] to track the desired trajectory for both the quadrotor and the payload. The electromagnet is controlled according to the planned pick and release times, t_m and t_r, in (40-f). To manage the change in weight resulting from picking and casting the payload, the

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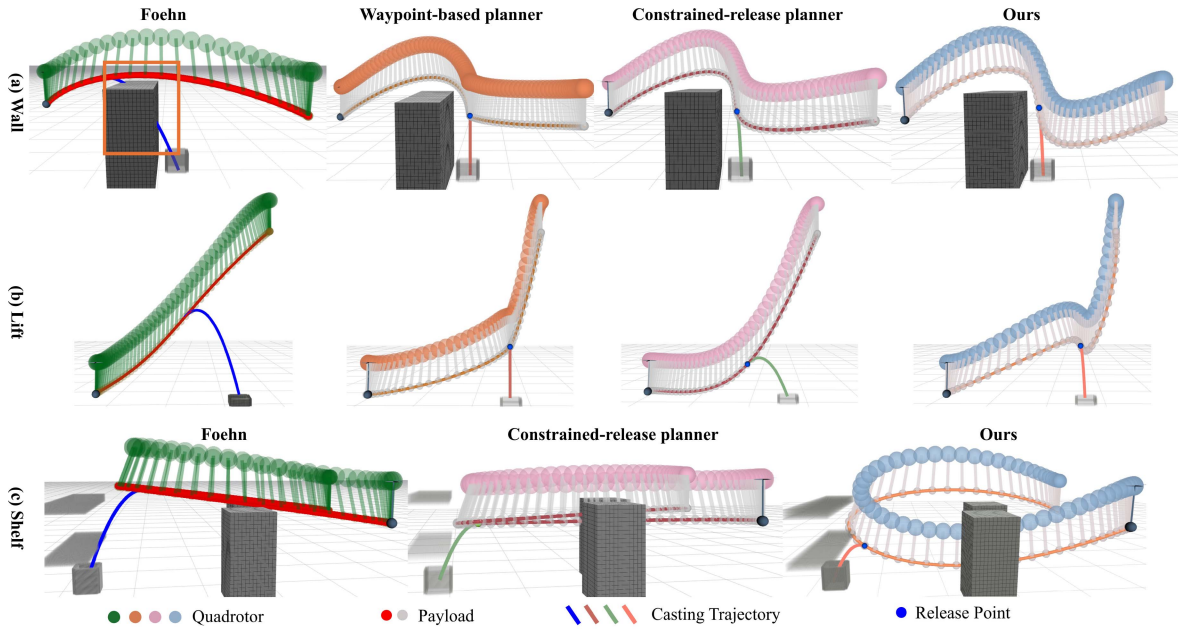


Fig. 10. Trajectory comparison in different scenarios. In scene (a) Trajectory from Foehn's method collides with obstacles because it lacks collision avoidance during the throwing motion. In scene (b) CR planner uses a high upward velocity, resulting in longer flight time. By contrast, our planner increases the downward release velocity to minimize airtime and improve accuracy, demonstrating the benefit of our error-bounded model. In scene (c) Our method finds a feasible path, while the WP method fails due to an obstructing shelf. Furthermore, the CR method's trajectories are less safe. They pass closer to obstacles and have a higher rate of collision because the method does not account for motion uncertainty.

payload mass is updated online as a parameter within the NMPC. The NMPC is implemented on an onboard computer and runs at a control frequency of 100 Hz.

VII. Simulation Evaluation

This section presents simulation studies designed to evaluate the proposed method under representative operating conditions. The evaluation addresses several questions within the tested regime, as follows.

1) How does our method compare against baseline approaches in terms of accuracy, safety, and efficiency (see Section VII-A)?
2) How sensitive is the planner to different levels of simulated uncertainty (see Section VII-B)?
3) What is the contribution of each component in the error-bounded model (see Section VII-C)?
4) How does the planner behave when using a different empirically estimated control-error bound (see Section VII-D)?
5) What is the impact of the hierarchical initialization strategy (see Sections VII-E and VII-F)?

Through quantitative analysis and ablation studies, we evaluate the effectiveness of the proposed approach.

A. Planning Performance Comparison

To quantitatively evaluate the performance of our planning method, we compare it against Foehn's method [4] and two baseline strategies: a waypoint-based planner (WP) and a constrained-release (CR). The Foehn's method optimizes for the nominal projectile motion without additional constraints, requiring the payload to land within a 5 cm range. We implement this using CasADi [44]. Both WP and CR baselines are derived from our optimization framework and share the same initialization strategy, but differ in their constraints are summarized as follows.

1) *WP:* This baseline removes uncertainty constraints and enforces a strict "hover-and-drop" strategy. It requires the system to come to a complete stop (zero velocity) directly above the target before releasing the payload.
2) *CR planner:* This baseline retains dynamic casting but replaces our adaptive error-bounded constraints with fixed, manually tuned limits on the release position and velocity. While it considers obstacle avoidance, it does not explicitly account for system uncertainty.

We evaluate the planners in the following three simulation scenarios, as depicted in Fig. 10.

1) *Scene (a) Wall:* The quadrotor must cast a payload into a box behind a wall.
2) *Scene (b) Lift:* The quadrotor lifts up and casts a payload onto the ground.
3) *Scene (c) Shelf:* The quadrotor casts a payload into a box under a shelf, avoiding two obstacles.

Our desired landing error is within 12 cm for scenes (a) and (b), and 15 cm for scene (c). For the baseline methods, the release distances for WP and CR planners are set to 1.0 m, 0.9 m, and 0.9 m for scenes (a), (b), and (c), respectively. The maximum release velocity for the CR planner is 2 m/s. Note that we set time penalty $\lambda_T = 500$ for our planner and baseline method.

1) Validation With Bounded Noise: First, to validate the effectiveness of our error bound constraint under a simple, bounded noise model, we inject a bounded control noise into the release states for each of the four methods in Scene (a). The noise is sampled from a uniform cylindrical distribution with magnitudes corresponding to the error bounds in Table I, where

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TABLE I
CONTROL ERROR BOUND VALUES

Control Error Bound	Value
Horizontal Position $\varepsilon_{\mathbf{p}}$ [cm]	$5.1 + 3.1 \left\|\ddot{\mathbf{x}}_L\right\|_2 - 0.42 \left\|\ddot{\mathbf{x}}_L\right\|_2^2$
Horizontal Velocity $\varepsilon_{\mathbf{v}}$ [cm/s]	15.8
Vertical Position ε_z [cm]	5.08
Vertical Velocity ε_{v_z} [cm/s]	5.0

TABLE II
CASTING ACCURACY WITH BOUNDED NOISE IN SCENE (A)

	Foehn	WP	CR	Ours
Max. Landing Error [cm]	17.9	13.1	12.2	**11.8**
Avg. Landing Error [cm]	6.1	4.8	4.4	**4.3**
Std. Landing Error [cm]	3.2	2.5	2.3	**2.2**

Bold indicates the best performance.

the horizontal position noise is set to 5.1 cm. For each method, we performed 10 000 simulations. The results, presented in Table II, demonstrate that our method successfully keeps the maximum landing error within the 12 cm objective while also achieving the lowest average landing error.

2) Comprehensive Comparison With Data-Driven Noise: To conduct a comprehensive performance comparison, we employ a realistic, data-driven noise model. Inspired by [45], we build a nonparametric probability distribution of control errors from over 30 000 empirical data points using kernel density estimation (KDE). KDE is a nonparametric method that estimates the probability density function of a random variable by summing symmetric kernel functions placed at each data point and smoothing them via a bandwidth parameter. This approach enables us to generate realistic error samples that preserve the data's original distribution. In our implementation, we use a Gaussian kernel with bandwidth $h = 0.1$. To capture the correlation between the system's state and control errors, we stratify the data into 1.0 m/s^2 bins based on the quadrotor's release acceleration and sample errors from the corresponding distribution. We perform 10 000 simulations for each planner across three scenarios to ensure a robust statistical comparison. The results, detailed in Table III, evaluate the planners on accuracy, safety, and efficiency. The in-range rate denotes the percentage of trials where the final landing error remains within the specified maximum allowable limit.

Accuracy: Our proposed method consistently achieves the highest casting accuracy across all scenarios with the lowest standard landing error. This is because it directly optimizes for landing accuracy under uncertainty. In contrast, the Foehn's method, which does not account for any term related to casting accuracy, yields the largest landing errors. While the WP can achieve comparable accuracy in open environments, its fixed release parameters (position, velocity, and acceleration) severely restrict the solution space. This limitation prevents the WP planner from finding feasible solutions in occluded environments, such as the Shelf scenario. In contrast, the larger solution space of our approach enables successful task completion in such complex cases. Furthermore, the WP method's requirement to stop at the release point (zero velocity) increases trajectory time. Our method avoids unnecessary stops by using dynamic casting maneuvers, thus improving time efficiency.

The CR planner lacks an accurate error model, and its accuracy degrades in some cases (e.g., the Lift scenario). A key insight is that simply reducing the release distance and velocity does not guarantee accuracy: flight time is equally critical, since longer airtime amplifies initial horizontal velocity errors. In the Lift scenario, CR releases at the same distance as our method but with a greater upward velocity, extending the flight time and increasing the landing error. By contrast, our planner increases the downward release velocity to minimize airtime (see Fig. 10), highlighting the benefit of our error-bounded model.

Furthermore, the CR method is highly sensitive to its manually tuned release parameters: too conservative increases trajectory duration, while too aggressive compromises accuracy. In contrast, our method directly sets the desired landing accuracy and, via the differentiable error-bounded model, adjusts the release states through gradient propagation to satisfy it. This yields feasible solutions within a larger solution space; for example, in the Wall scenario, our planner achieves better accuracy than CR even with a larger release distance.

Safety: Our planner improves safety by enforcing collision avoidance for the payload's uncertain ballistic trajectory, a clear advantage over Foehn's method, which ignores projectile collisions entirely (see Fig. 10, Wall). Even compared to a planner with nominal obstacle avoidance (CR), our error-bounded method performs better: in the challenging shelf scenario, it reduces the collision rate from 39.7% (CR) to 2.5%, showing that accounting for uncertainty is crucial for safety in cluttered environments.

Computational efficiency: Our method is also significantly more computationally efficient than Foehn's due to our compact problem formulation and hierarchical initialization. Unlike the Foehn's method, which optimizes over the full state trajectory, our approach leverages differential flatness to drastically reduce the number of decision variables. This simplification, combined with our hierarchical initialization strategy that provides a high-quality initial guess, allows our planner to converge approximately 10 times faster than Foehn's method. Moreover, despite incorporating the error-bounded model, our method's computation time does not increase significantly compared to the simpler WP and CR planners, demonstrating that our error-bounded constraints are highly efficient (see Section VII-F for an ablation on the initialization).

B. Sensitivity to Different Levels of Uncertainty

In this section, we analyze the casting accuracy with respect to the level of uncertainty in the payload's state. To do this, we varied the magnitude of simulated noise applied to the payload's position and velocity, scaling it from 10% to 120%. The standard magnitude of noise is defined in Table I. The trajectory was kept the same. We evaluated the resulting maximum landing error and compared our planner's performance against the Foehn's method [4] within the "Lift" simulation scenario. The desired maximum landing error is 12 cm at the 100% noise level. Note that in this experiment, we use bounded noise, and the

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TABLE III
PLANNING PERFORMANCE COMPARISON

Case	Wall				Lift				Shelf			
Method	Foehn	WP	CR	Ours	Foehn	WP	CR	Ours	Foehn	WP	CR	Ours
Max. Landing Error [cm]	23.0	14.1	13.8	**13.2**	24.8	13.9	25.2	**13.2**	31.8	-	30.5	**19.2**
Avg. Landing Error [cm]	6.9	3.0	3.0	**2.8**	7.2	3.1	8.7	**2.9**	9.4	-	8.4	**6.5**
Std. Landing Error [cm]	4.6	2.5	2.5	**2.3**	4.9	2.4	4.9	**2.3**	5.2	-	5.1	**4.0**
In-Range Rate [%]	88.67	99.85	99.89	**99.94**	87.16	99.89	79.93	**99.97**	86.15	-	92.14	**99.31**
Load Flight Time [s]	0.55	0.46	0.42	0.38	0.72	0.43	0.56	0.39	0.56	-	0.4	0.36
Release Distance [m]	1.9	1.0	1.0	1.1	1.9	0.9	0.9	0.9	2.0	-	0.9	0.7
Release Vel. [m/s]	2.1	0.0	0.7	1.1	1.6	0.0	2.0	0.5	2.1	-	1.2	1.5
Release Acc. [m/s^2]	0.8	0.0	0.3	0.0	1.0	0.0	2.0	0.0	2.4	-	2.8	2.2
Landing Vel. [m/s]	5.9	4.4	4.5	4.7	5.9	4.2	4.1	4.2	5.6	-	4.1	3.6
Casting Safety	No	No	Nominal	**Yes**	No	No	Nominal	**Yes**	No	No	Nominal	**Yes**
Casting Collision Rate [%]	100.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	71.7	-	39.7	**2.5**
Traj. Duration [s]	**5.39**	6.71	5.60	5.52	4.40	4.99	**4.21**	4.79	6.65	-	6.59	**6.54**
Comp. Time [s]	2.69	0.089	**0.053**	0.058	1.25	**0.105**	0.115	0.120	5.48	-	0.099	**0.090**

Bold indicates the best performance.

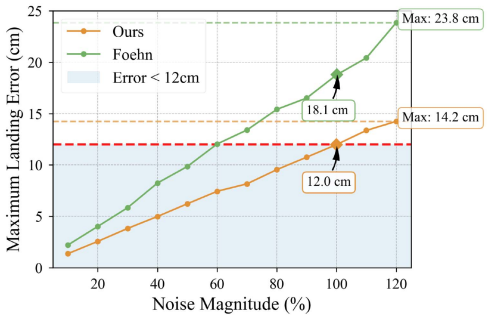


Fig. 11. Comparison of casting accuracy with respect to control noise. The plot compares the maximum landing error of our method against Foehn's method [4] as the magnitude of simulated control noise is scaled. The landing error of our approach increases more modestly than that of Foehn's method, as indicated by the gentler slope of its corresponding curve.

TABLE IV
ABLATION STUDY ON CASTING ACCURACY

	Nominal	w/o Horiz.	w/o Vert.	Full
Max. Landing Error [cm]	36.1	31.9	15.1	**13.2**
Avg. Landing Error [cm]	13.4	12.2	5.3	**2.9**
Std. Landing Error [cm]	6.4	5.9	2.7	**2.3**
Success Rate [%]	43.91	51.47	99.40	**99.97**

Bold indicates the best performance.

TABLE V
ABLATION STUDY ON LANDING VELOCITY ERROR BOUNDS

	w/o Horiz.	w/o Vert.	Full
Vertical Violation Rate [%]	0.01	18.5	**0.0**
Horizontal Violation Rate [%]	16.2	0.0	**0.0**

Bold indicates the best performance.

noise vector is sampled from a uniform cylindrical distribution centered at the mean state value.

The results in Fig. 11 indicate that our method is less sensitive to increasing control error: its maximum landing error grows more slowly with noise magnitude (i.e., a gentler slope) than Foehn's method. For example, when increasing the noise scale from 100% to 120%, our error increases by 2.2 cm and the error-growth rate is 18%, whereas Foehn's method increases by 5.7 cm and the error-growth rate is 31%, reflecting a larger error-growth rate.

C. Ablation Study on Error-Bounded Model

This ablation study is conducted to analyze the contribution of each control error bound. To ensure the realistic performance comparison, we utilize the data-driven noise model in the ablation study, which is the same as in Section VII-A2.

1) Casting Accuracy: We conduct an ablation study on casting accuracy to analyze the contribution of each control error bound. We compare four variations of our planner: one without any control error bounds (nominal), one without the horizontal error bound term (w/o Horiz.), one without the vertical error bound term (w/o Vert.), and our proposed method which considers all error bounds (full). The analysis is conducted in the lift scenario.

The results are presented in Table IV. Our full model demonstrates superior performance across all metrics, achieving an average landing error of only 2.9 cm and a success rate of 99.97%. The nominal model, which ignores all control error, performs the worst, with a maximum error of 36.1 cm. Notably, omitting horizontal error has a more significant impact on accuracy than omitting vertical error. Nevertheless, only the comprehensive model that accounts for all error bounds can achieve the highest level of accuracy and robustness.

2) Landing Velocity: We perform another ablation study to investigate the impact of different error bounds in the landing velocity constraint. In all test cases, the maximum horizontal and vertical landing velocities are constrained to 1.5 m/s and 3.8 m/s, respectively. We only vary whether the planner accounts for the control error in the constraint. w/o Horiz. and w/o Vert. mean neglecting the control error in the horizontal and vertical directions, respectively.

The results, summarized in Table V, show that neglecting control error in either direction makes the planner overly optimistic.

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TABLE VI
ABLATION STUDY ON PICKING ACCURACY

	Nominal	w/o Horiz.	w/o Vert.	Full
Success Rate [%]	15.42	24.78	58.24	**97.80**
Horizontal Failure [%]	79.29	75.22	2.20	**2.20**
Vertical Failure [%]	27.93	0.00	40.08	**0.00**

Bold indicates the best performance.

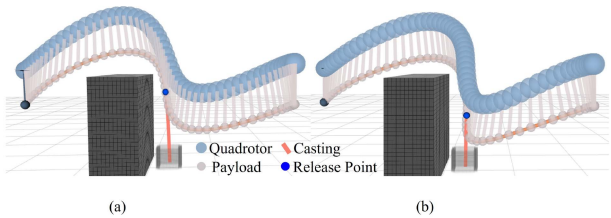


Fig. 12. Trajectory comparison for different control error bounds. (a) Trajectory with the original error model. (b) Trajectory with the larger error model. With a larger control error, our planner adaptively adjusts the payload's release point to guarantee landing accuracy.

Under the influence of control noise, these solutions frequently violate the landing velocity constraints. In contrast, our full model satisfies both constraints, highlighting the necessity of each error bound.

3) Picking Accuracy: Finally, we evaluate the influence of error bounds on picking accuracy. A picking attempt is deemed successful if the quadrotor's electromagnet is horizontally within 6 cm of the payload's location and positioned no more than 2 mm above its top surface. In our simulation, this threshold represents the effective engagement range of the electromagnet, ensuring the magnetic force is sufficient to capture the payload.

As shown in Table VI, our full error-bounded model achieves a remarkable 97.80% success rate, significantly outperforming all other variants. The nominal planner's success rate drops to 15.42%. These results underscore that a comprehensive error-bounded model, addressing both horizontal and vertical components, is indispensable for achieving reliable payload picking.

D. Ablation Study on Different Control Error Bounds

This experiment investigates the adaptive capabilities of our planner under different control error bounds. To induce a larger control error, we used a heavier payload (148 g) compared to the original one (84 g). We collected new control error data with the heavier payload from the real system and constructed a corresponding nonparametric empirical noise model for simulation as presented in Section VII-A2. For the heavier payload, the error bounds were empirically updated to: horizontal position $\varepsilon_{\mathbf{p}} = 5.3 + 4.0\|\ddot{\mathbf{x}}_L\|_2 - 0.38\|\ddot{\mathbf{x}}_L\|_2^2$ cm, horizontal velocity $\varepsilon_{\mathbf{v}} = 20.1$ cm/s, vertical position $\varepsilon_z = 7.2$ cm, and vertical velocity $\varepsilon_{v_z} = 5.5$ cm/s. This experiment evaluates whether our planner can maintain landing accuracy when faced with increased control uncertainty.

As illustrated in Fig. 12, the planner adapts its strategy in response to the increased uncertainty. The heavier payload results in a larger horizontal velocity error. Since a longer flight time would amplify this velocity error and degrade landing accuracy, the planner strategically lowers the release point. This adjustment reduces the payload's time in the air, thereby mitigating the effect of the larger initial error.

TABLE VII
CASTING ACCURACY WITH DIFFERENT CONTROL ERROR BOUNDS

	Max. Error [cm]	Avg. Error [cm]	In-Range Rate [%]
Original	13.2	2.8	99.94
Heavy	13.4	2.8	99.78

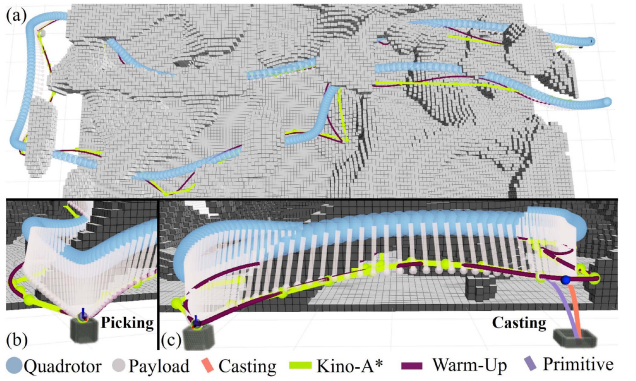


Fig. 13. Trajectory comparison for the ablation study on joint optimization. The figure displays trajectories from different stages of our hierarchical planner in a complex Perlin environment: the initial path from Kinodynamic A* (green), the trajectory after warm-up optimization (purple), and the final jointly optimized trajectory (gray). The final trajectory is visibly smoother and more direct. This showcases the benefits of our joint optimization.

To quantitatively validate this adaptation, we performed 10 000 casting simulations for both error models in the Wall scenario. The results, summarized in Table VII, confirm that the planner's adjustments are effective. Despite the larger control errors associated with the heavier payload, the landing accuracy is maintained. The maximum and average landing errors remain almost identical to the original case, and the in-range rate shows only a marginal decrease of 0.16%.

These results show that our planner can be adapted to different system configurations and control performance levels by updating the error bounds accordingly.

E. Ablation Study on Joint Optimization

To demonstrate the effectiveness of our joint optimization strategy, we compare the final trajectory with intermediate results from our hierarchical planning pipeline. The experiment is performed in a complex Perlin noise environment, as depicted in Fig. 13. The environment size is 15 m $\times$ 15 m, providing a challenging setting with irregular obstacles. We analyze three stages of the trajectory generation process: only Kinodynamic A* (Kino-A*), adding a warm-up optimization (Warm-up), and our full joint optimization (Joint Opt.).

The quantitative results are summarized in Table VIII. Joint optimization yields a significant improvement in both trajectory duration and length. Compared to the initial Kino-A* path, the final trajectory's duration is reduced by 42.5%, and its length is reduced by 7.7%. The warm-up stage provides a notable improvement over the initial path. However, only the joint optimization finds the most efficient solution. It achieves this by seamlessly integrating all phases of the delivery task.

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TABLE VIII
ABLATION STUDY ON JOINT OPTIMIZATION

	Kino-A*	Warm-up	Joint Opt.
Time [s]	37.5	29.1	**21.5**
Trajectory Length [m]	47.1	46.2	**43.5**

Bold indicates the best performance.

TABLE IX
COMPUTATIONAL EFFICIENCY COMPARISON (TIME IN MS)

	Strategy	T_{init}	T_{warm}	T_{opt}	Total
(a) Occ.	1) Hierarchical Init.	1.2	20.7	67.7	**89.6**
	2) w/o Warm-Up	1.2	-	119.0	120.2
	3) w/o Kino-A*, Warm-Up	0.6	-	257.5	258.1
	4) Straight Line	-	-	-	✗
(b) Perlin	1) Hierarchical Init.	104	175	131	**410**
	2) w/o Warm-Up	104	-	383	487
	3) w/o Kino-A*, Warm-Up	-	-	-	✗
	4) Straight Line	-	-	-	✗

Note: T_{init}, T_{warm}, and T_{opt} denote the time consumption for Initial Search, Warm-Up Optimization, and Final Trajectory Optimization, respectively. Units: milliseconds. Bold indicates the best performance.

Furthermore, the final trajectory satisfies all feasibility constraints. The maximum velocity and acceleration of the payload are 3.0 m/s and 2.0 m/s^2, respectively. The minimum cable tension throughout the trajectory is 1.03 N.

F. Ablation Study on Computational Efficiency

In order to investigate the computational efficiency of our planning method, we conduct an ablation study on the hierarchical initialization strategy. Two scenarios are considered: (a) occluded casting, and (b) picking and casting in a Perlin environment. The first scenario replicates the real-world experiment described in Section VIII-B, where we emphasize the computational efficiency performance under strict casting constraints. The second scenario corresponds to the Perlin environment shown in Fig. 13, where we evaluate our method in a larger, more intricate setting. We compare the following four different initialization strategies.

1) *Full hierarchical initialization:* Our proposed method, which integrates casting primitive selection, iterative Kinodynamic A*, and warm-up optimization.
2) *Hierarchical initialization without warm-up:* Same as strategy (1) but without the warm-up optimization.
3) *Primitive-based initialization without kinodynamic A* and warm-up:* Using a straight line for initialization while still preserving the casting primitive selection.
4) *Straight line initialization:* A direct connection of the start, picking target, casting target, and endpoint is used as the initialization.

We record the running time of the algorithm under each strategy, and the corresponding results are summarized in Table IX. The total time represents the whole planning pipeline runtime. From our study, three conclusions can be drawn:

First, comparing strategies (2) and (3) reveals that directly optimizing the entire trajectory without warm-up initialization requires more time to converge. The incorporation of warm-up optimization reduces the planning time by 30% in the occluded casting scenario and by 16% in the Perlin scenario. The presence of complex casting and picking constraints can increase the number of optimization iterations, but warm-up optimization helps guide the process toward a suitable local minimum, thereby speeding up convergence.

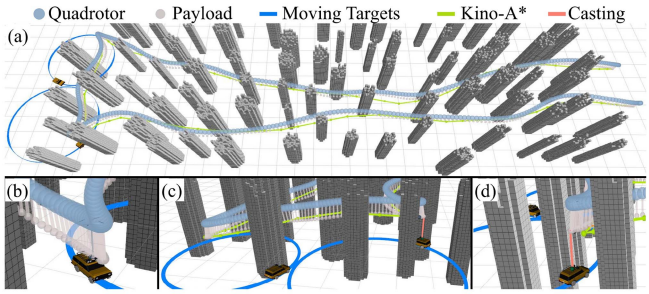


Fig. 14. Dynamic picking and casting in a 10 m × 30 m random cylinder environment. (a) Overview of the generated trajectory (approx. 75 m in length). (b) and (d) Detailed views of the picking and casting phases, respectively. (c) Trajectory segment transitioning between picking and casting. The moving platforms follow circular trajectories, depicted as blue path. The quadrotor successfully navigates the dense obstacles to transfer the payload between the moving platforms.

Second, the casting primitive selection is essential when the casting trajectory must avoid obstacles. In scenario (a), omitting the casting primitive [strategy (4)] results in the optimizer failing to converge because the casting obstacle avoidance constraint can trap the optimization process. A proper initialization provided by the casting primitive selection makes the overall optimization more tractable. Nevertheless, this strategy may fail if the casting primitives are too sparse to yield a feasible solution in occluded environments. We address this by increasing the primitive density, using a 10 cm grid spacing in our experiments.

Third, in complex obstacle environments, as indicated by the difference between strategies (2) and (3), Kinodynamic A* plays a crucial role. A straight line is insufficient for initialization in Perlin environment. During the search process, Kinodynamic A* not only provides a collision-free path but also assigns a reasonable duration to each segment, which aids in better initializing the trajectory duration in the optimization. This results in fewer iterations and faster overall convergence.

In conclusion, the hierarchical initialization strategy achieves the lowest computation time in both scenarios, significantly enhancing computational efficiency.

G. Qualitative Evaluation

1) Dynamic Pick-and-Cast: We evaluate our method in a dynamic pick-and-cast scenario, as shown in Fig. 14, where the quadrotor retrieves a payload from one moving vehicle and casts it to another. This experiment is conducted in a large-scale random cylinder environment to simulate a forest-like setting. This task highlights our planner's capability to execute dynamic picking and casting within a cluttered, complex environment.

2) Ablation Study on Object-Centric ESDF: To qualitatively demonstrate the effectiveness of the object-centric ESDF, we perform an ablation study in a dynamic pick-and-cast scenario, as depicted in Fig. 15. The primary challenge is preventing the payload from colliding with the moving platform from which

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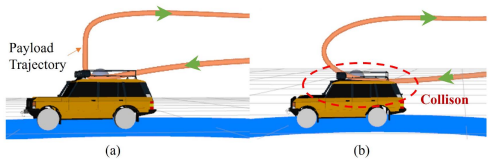


Fig. 15. Ablation study of the object-centric ESDF. (a) With the object-centric ESDF, the planner generates a safe trajectory that successfully avoids the picking platform. (b) Without it, the payload collides with the platform from which it was retrieved.

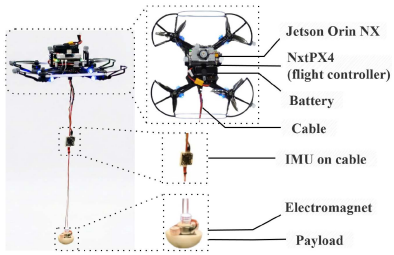


Fig. 16. Quadrotor-payload platform. The quadrotor is equipped with an electromagnet to hold the payload. The payload is attached to the quadrotor by a cable.

it was just retrieved. As shown in the comparison, a planner without the object-centric ESDF fails, causing the payload to collide with the platform. In contrast, our method generates a safe and feasible trajectory, successfully avoiding the obstacle.

VIII. Real-World Experiments

A. System Setup

We have developed a quadrotor-payload platform that utilizes an electromagnet to hold a payload, enabling stable flight and precise delivery and casting. The platform is shown in Fig. 16. Our platform incorporates a 250 mm frame equipped with NxtPX4 [46], a high-performance mini flight control system compatible with PX4. For computing capabilities, we have employed the NVIDIA Jetson Orin NX, which serves as the platform for executing all algorithms. The payload release mechanism is actuated by the electromagnet, and the payload is tethered by a cable of length 53 cm. An additional inertial measurement unit is affixed to the cable to estimate its direction and the payload's state, providing real-time feedback for accurate trajectory tracking. The position of the quadrotor is provided by a motion capture system.

B. Casting to Static Target

To demonstrate the capability of our planning framework for adaptive and accurate payload casting, we evaluate its performance in a real-world occluded casting scenario within a static environment (see Fig. 17). In this experiment, a quadrotor was tasked with navigating through obstacles to deliver a payload into a 30 cm $\times$ 30 cm box located beneath a shelf. The gap between the shelf and the box was approximately 30 cm, creating a highly constrained environment. This setup demonstrates that our framework can ensure safe and successful payload delivery even under such challenging conditions. Additional validation experiments are available in the supplementary video.

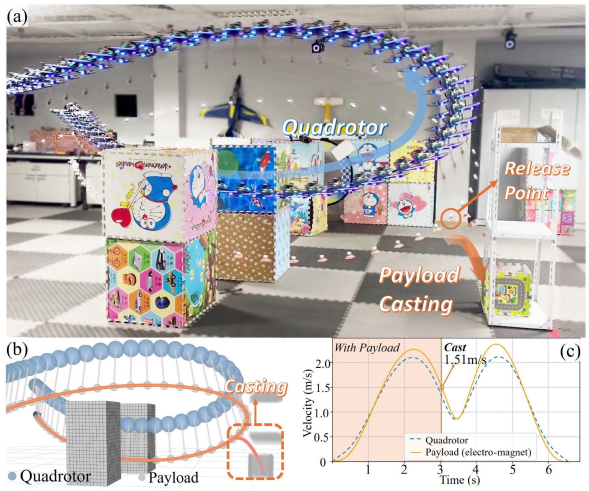


Fig. 17. *Real-world Exp: Occluded casting*: Our planner can generate a maneuver to cast the payload into the box occluded by the shelf. (b) and (c) Planned trajectory and velocity.

C. Casting to Moving Platform

We evaluate the planner's ability to cast a payload to a moving target. In the first experiment, the target follows a known eight-shaped trajectory, and the quadrotor successfully delivers the payload in this predictable dynamic setting. To further evaluate real-time replanning under target-motion uncertainty, a human operator manually drives the platform, so the quadrotor has no prior knowledge of its trajectory and only receives real-time position and velocity, predicting future motion with a constant-velocity model. By replanning at 10 Hz, the planner still achieves safe and accurate delivery under the unpredictable motion. Full demonstrations are provided in the accompanying video.

D. Delivery With Picking and Casting

We validate the full delivery pipeline with two real-world experiments. In the first, the quadrotor picks a payload from a static box and casts it into a moving basket whose nominal trajectory is known, with two obstacles added to the scene. Please see the video for a full demonstration.

In the second experiment, the quadrotor picks up a payload from a moving platform and casts it into a moving basket, verifying the planner's performance with dynamic targets. The moving platform follows a circular trajectory, and the moving basket follows an eight-shaped trajectory. Both maximum velocities are set to 0.75 m/s. The quadrotor is informed of the moving platform's nominal trajectories and can predict their expected position and velocity. The trajectory's uncertainty is modeled as a constant bound.

As shown in Fig. 18, the quadrotor synchronizes its motion with the moving platform for picking at 0.75 m/s and then casts into the basket. Despite the noisy velocities of the moving targets [see Fig. 18(d)], the planner accounts for this uncertainty to ensure robust picking and casting, successfully delivering the payload in this tested scenario.

These experiments demonstrate that our method enables seamless, nonstop execution of both picking and casting, thereby improving the overall time efficiency of the delivery.

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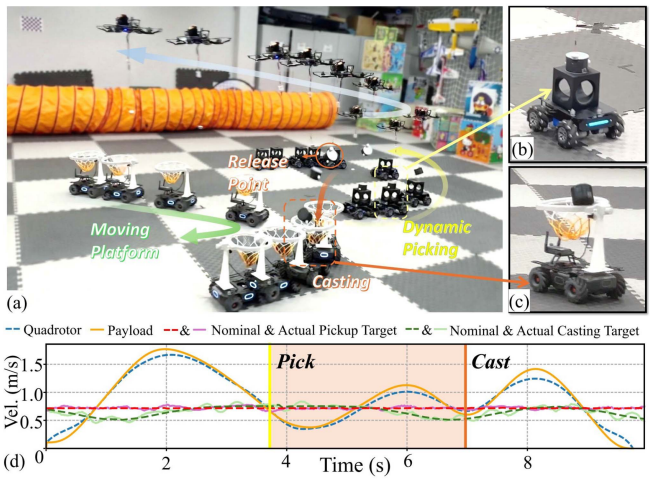


Fig. 18. *Real-world Exp: Delivery Case*: The quadrotor picks the payload from a moving platform and casts it into a moving basket. (b) and (c) Highlight the picking and casting processes, respectively. (d) Velocity profiles of the quadrotor, payload, and both moving targets. The actual velocities of picking and casting targets are noisy. Despite the noise, the planner accounts for the uncertainty to ensure accurate and safe delivery.

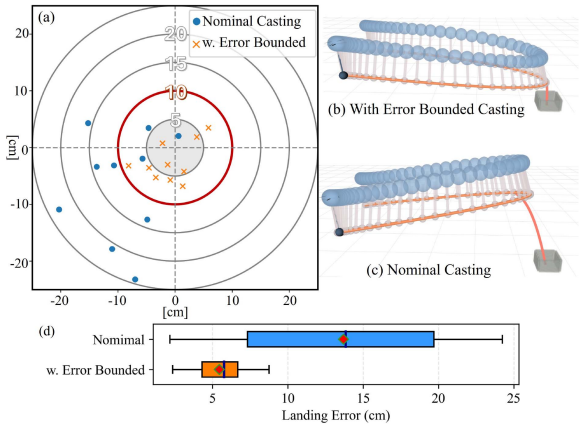


Fig. 19. *Ablation study on casting accuracy*: (a) Payload's landing positions in a real-world experiment. The red circle indicates the maximum acceptable error. (b) and (c) Compare the trajectories generated with and without error-bounded casting, respectively. (d) Boxplot of landing errors. The boxes represent the interquartile range, while the horizontal line and the diamond inside each box denote the median and mean value, respectively.

E. Casting Accuracy Comparison

To evaluate casting performance in real-world conditions, we conducted an experiment comparing nominal casting with error-bounded casting. In nominal casting, the release position and velocity are assumed to be accurate, and the projectile is assumed to reach the target point directly without considering landing accuracy. In contrast, error-bounded casting accounts for uncertainties in the payload's release position and velocity, resulting in a more conservative and robust trajectory.

We perform 10 trials for each approach in a real-world setting. The landing positions are shown in Fig. 19(a). The red circle represents the maximum acceptable error, which we set as 10 cm. All trials using error-bounded casting fall within this range, whereas nominal casting exhibits larger deviations. As detailed in Table X, the proposed method reduces the average and maximum landing errors by 60% and 64%, respectively. Crucially, the 95% CI for the two methods do not overlap, and the Hedges' g effect size for small samples [47] is 1.43 (> 0.8), confirming a large and statistically significant improvement. The boxplot in Fig. 19(d) further shows that our method significantly lowers the range of landing errors compared to the nominal baseline.

TABLE X
LANDING ACCURACY COMPARISON IN REAL ENVIRONMENT

Method	Mean [cm]	95% CI [cm]	Std [cm]	Max [cm]
Nominal	13.69	[8.27, 19.11]	7.19	24.2
w. Error-bounded	**5.46**	**[4.09, 6.82]**	**1.81**	**8.75**

Bold indicates the best performance.

IX. CONCLUSION

A. Choice of Control Error Model

We use a deterministic, data-driven bounded-error model rather than a stochastic Gaussian model for two reasons. First, the control error does not follow a Gaussian distribution. We conducted normality tests on the collected data from real-world control experiments. Specifically, we performed four distinct normality tests: Shapiro–Wilk, D'Agostino-Pearson, Anderson–Darling, and Kolmogorov–Smirnov. All tests consistently rejected the normality hypothesis for control errors with high confidence ($p < 0.001$). Our distribution-free formulation avoids relying on an inaccurate parametric error model.

Second, nonlinear propagation yields non-Gaussian landing errors. Even if the control errors were Gaussian, the nonlinear casting process produces a landing distribution that is neither Gaussian nor isotropic, with no accurate analytic expression. Sampling-based methods, such as Sample Average Approximation could approximate it but are computationally intensive, requiring thousands of simulated trajectories. By propagating only the error bounds, our method can efficiently computes a guaranteed landing-error bound without relying on any specific distribution.

B. Limitations

Although the proposed method is evaluated through simulation experiments, current real-world experiments are conducted indoors on a single quadrotor-payload platform with motion-capture localization. Due to practical constraints of the experimental setup and indoor venue, the real-world tests could not cover a broader range of scenarios, such as outdoor operation with wind disturbances, larger workspaces, and diverse payload configurations.

For dynamic targets, the current planner predicts target motion using a constant-velocity model and replans at 10 Hz. This strategy is effective for the tested target motions, but may fail when the target performs aggressive maneuvers, erratic motion, or sudden stops that violate the assumed error bounds. In such cases, the payload may miss the target or produce landing errors outside the prescribed bounds. More sophisticated target prediction models and adaptive replanning strategies are needed to mitigate these failure cases.

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The modeling assumptions also introduce limitations. Air resistance is neglected to preserve the differential-flatness-based planning formulation and avoid solving transcendental equations for the casting time of flight. This approximation is reasonable for the low-speed indoor experiments considered in this work, but may become less accurate for high-speed casting or lightweight payloads. In addition, the current control-error bound model is approximate: the horizontal position bound is represented by an acceleration-dependent second-order polynomial, while the remaining bounds are treated as constants estimated from empirical data. Although this compact model balances fidelity and computational efficiency, it may not capture higher order dependencies on velocity, attitude, or payload properties. Future work will broaden the experimental validation and investigate data-driven error models, together with more systematic sensitivity and failure-case analyzes.

C. Conclusion

This article presents a unified planning framework for autonomous delivery with a quadrotor-payload system, addressing the core challenges of accuracy, flexibility, and computational efficiency. We introduce an error-bounded method to constrain picking and casting errors under a bounded-error uncertainty model, a compact optimization formulation to unify the delivery process, and a hierarchical initialization strategy to improve computational performance. Simulation ablations and indoor real-world experiments validate the effectiveness of the proposed method within the tested operating conditions.

APPENDIX

A. Convexity of the Landing Error Bound

In Section IV, we state that the worst-case landing error, $E(\tilde{t}_f) = \|\mathbf{a} + \mathbf{b}\tilde{t}_f\|_2 + c + d\tilde{t}_f$, is a convex function with respect to the flight time $\tilde{t}_f$. The $\mathbf{a} = \mathbf{p}_r^{xy} - \mathbf{p}_{d|t_r+t_f} + \dot{\mathbf{p}}_{d|t_r+t_f} t_f$, $\mathbf{b} = \mathbf{v}_r^{xy} - \dot{\mathbf{p}}_{d|t_r+t_f}$, $c = \varepsilon_{\mathbf{p}} + \varepsilon_{\mathbf{p}_d}$, and $d = \varepsilon_{\mathbf{v}}$ are constants with respect to $\tilde{t}_f$. The first term, $\|\mathbf{a} + \mathbf{b}\tilde{t}_f\|_2$, is a composition of the Euclidean norm (a convex function) and an affine function of $\tilde{t}_f$, which results in a convex function. The second term, $c + d\tilde{t}_f$, is an affine function and is, therefore, convex. Since the sum of convex functions is convex, $E(\tilde{t}_f)$ is also convex.

B. Monotonicity of the Payload's Flight Time

The payload's flight time $\tilde{t}_f$, as given by (6), increases monotonically with respect to the vertical release deviations ξ_z and ξ_{v_z}. We prove this by examining its partial derivatives. For position deviation the derivative $\frac{\partial \tilde{t}_f}{\partial \xi_z} = [(\dot{z}_r + \xi_{v_z})^2 + 2g(z_r - z_g + \xi_z)]^{-\frac{1}{2}}$ is strictly positive.

For velocity deviation ξ_{v_z}, let $\tilde{v}_z = \dot{z}_r + \xi_{v_z}$. The derivative is $\frac{\partial \tilde{t}_f}{\partial \xi_{v_z}} = \frac{1}{g}\left(1 + \frac{\tilde{v}_z}{\sqrt{\tilde{v}_z^2 + 2g\Delta h}}\right)$. Since the release height $\Delta h \geq 0$, the denominator $\sqrt{\tilde{v}_z^2 + 2g\Delta h} \geq |\tilde{v}_z|$. This implies the fraction is within $[-1, 1]$, making $\frac{\partial \tilde{t}_f}{\partial \xi_{v_z}} \geq 0$. Thus, $\tilde{t}_f$ is nondecreasing with respect to both deviations.

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